

Hadronic Shower Characterization in the Belle II Electromagnetic Calorimeter

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Abstract

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0 Chapters subdivision

The anti-neutron in physics: the Chapter 1 describes the properties of the anti-neutron and its detection techniques in the high-energy physics (HEP) context. Its primary interactions -such as annihilation and hadronic shower production- are then exploited, discussing some hadronic models very quickly. The MANTRA (Measuring Anti-Neutron: Tagging and Reconstruction Algorithm) project is then described as a method to measure anti-neutron energies up to 10 GeV using an electromagnetic calorimeter.

The Belle II experiment: the Chapter 2 provides an overview of the SuperKEKB accelerator and Belle II detector. In the context of $\bar{n}$ studies, the Electromagnetic Calorimeter (ECL) plays a key role in hadron detection; a detailed discussion about the ECL is thus provided, covering the description of its inorganic crystals and the explanation of the clustering algorithm. The ECL performance are then discussed.

Then the Belle II Analysis Software Framework (basf2) is introduced, providing an overview of its operation method and its role in the Monte Carlo simulation and reconstruction processes at Belle II experiment. Afterwards, the Topology Analysis (TopoAna) independent software is described.

Study of the signal sample channel: the Chapter 3 is devoted to the study of the kinematic variables of anti-neutrons $\bar{n}$, including momentum, angular distributions and related observables. The analysis follows a semi-exclusive approach over the selected channel $e^+e^- \rightarrow p\pi^-\bar{n}$, studying both cases with and without Initial State Radiation (ISR) reconstruction, where the recoil system is made of $p\pi^-(\gamma_{ISR})$. Therefore, the recoil system variables are compared to those of the anti-neutrons candidates, studying the response of a Kinematic Fit as well.

Cluster split-off and pre-showering effects are investigated using studying anti-neutrons relatives chain. Several neutral cluster variables associated to anti-neutron candidates are then described and studied for both ISR and NO ISR cases, including the analysis of their relatives contributions.

1 Introduction

The Standard Model is one of the greatest achievements of modern science, especially in the field of high-energy particle physics. Its reliability is the result of a long process that began in the 20th century and received its definitive confirmation in 2012 with the discovery of the Higgs boson.

However, there are some topics which remain unsolved, like the non-trivial neutrino mass or the presence of dark matter on cosmological scales. The Standard Model still represents an excellent phenomenological description of subatomic processes, at the $\lesssim 1$ TeV energy scale.

In this framework, the **Belle II** experiment, located at the SuperKEKB facility, plays a fundamental role in exploring the aforementioned topics. These are scenarios that go beyond the standard physics, thus entering the so-called *new physics*.

In parallel with experiments that maximize the energy, the aim of Belle II is to explore the intensity frontier. Its main goal is not to increase the center-of-mass energy, but rather to explore scenarios of high instantaneous luminosity up to $8 \times 10^{35} \text{ cm}^{-2} \text{ s}^{-1}$, around (and not limited to) the $\Upsilon(4S)$ resonance energy.

This approach enables flavour physics studies of heavy mesons such as B mesons, facilitating the identification of potential discrepancies with Standard Model predictions and opening the door to Beyond Standard Model physics.

1.1 High Luminosity B Factory

Since the end of the XX century, two asymmetric-energy e^+e^- B factories - KEKB for the Belle experiment at KEK and PEP-II for the BaBar experiment at SLAC - have achieved huge success, confirming the Standard Model in the quark flavour sector, establishing the presence of CP violation in the B meson system through the decay process $B^0(\bar{B}^0) \rightarrow J/\Psi K_S^0$ [6, 7, 8, 9]. This result fixed the main target of the present B factories. Experimental results confirm that the Kobayashi–Maskawa mechanism, a key component of the Standard Model of elementary particles, is the dominant source of CP violation observed in nature.

The Belle experiment proved its ability to measure a number of decay modes of the B meson and to extract Cabibbo-Kobayashi-Maskawa (CKM) matrix elements and other interesting observables. Precise measurements and new observations from Belle have overconstrained the quark flavor sector of the Standard Model. This sector remains self-consistent within the current experimental precision.

However, several fundamental questions persist in the quark and lepton flavor sectors; the Standard Model requires numerous parameters, including quark and lepton masses and mixing angles that must be determined experimentally.

Furthermore, from a cosmological perspective, the matter-antimatter asymmetry of the Universe poses a great challenge. While the CP violation constitutes one of the necessary conditions for the evolution of a matter-dominated Universe [10], its magnitude cannot be explained solely by the CP violation within

the Standard Model, which originates from quark mixing.

Exploring such scenarios requires substantial luminosity at present B factories. The SuperKEKB Higher Luminosity B Factory has been designed to investigate the physics of B mesons processes. Due to the very large production cross section of b -quarks in hadron collisions, certain measurements such as $B_s \rightarrow \mu\mu$ can be performed more effectively at hadron colliders, such as LHC. However, e^+e^- machines provide a much cleaner environment, essential for precision observables involving photons, π^0 's, K_L^0 's, or neutrinos in the final state.

At the $\Upsilon(4S)$ resonance, $B\bar{B}$ pairs are produced near threshold without associated particles. By fully reconstructing the four-momentum of one B ($\bar{B}$) meson from its decay products (full reconstruction technique), the missing momentum of the other $\bar{B}$ (B) meson decay can be inferred. This technique is essential for measuring channels with neutrinos in the final state. Therefore, SuperKEKB undoubtedly complements hadron collider experiments from both theoretical motivation and experimental perspectives.

1.2 Anti-neutron in HEP physics

The anti-neutron $\bar{n}$ ($\bar{u}\bar{d}\bar{d}$) is the anti-particle of the neutron n (udd), a neutral hadron of mass of around $0.939565 \text{ GeV}/c^2$ and spin $\frac{1}{2}$, with magnetic moment of $1.91\mu_N$, where μ_N is the nuclear magneton. Its existence was expected by the principle of invariance under charge conjugation and experimentally proved in 1956 [11], one year after the anti-proton $\bar{p}$ [12].

The anti-neutron $\bar{n}$ plays a key role in several physics measurements, both in collider physics and in astrophysical contexts. Since it is electrically neutral, it cannot be easily observed directly, allowing its detection via the products of its annihilation with ordinary matter.

For example, the measurement of the neutron electromagnetic form factor in the time-like region is strongly affected by the performance of the $\bar{n}$ detection. The BESIII experiment has shown [13] an oscillating behaviour in the electromagnetic structure of the neutron in the study of the $e^+e^- \rightarrow n\bar{n}$ annihilation process, where the two neutral hadrons in the final state were reconstructed using the well-known two-body kinematics. Direct detection of the $\bar{n}$ would further refine this analysis by enabling a more proper identification of the final state.

Another example, where an improved direct measurement of the $\bar{n}$ would significantly enhance the studies, is the analysis of hyperon decay channels at B-factories (and beyond). Some possible decay channels where the $\bar{n}$ is the only un-reconstructed particle in the final state are shown below:

$$\bar{\Lambda}^0 \rightarrow \pi^0 + \bar{n}, \quad \bar{\Sigma}^- \rightarrow \pi^- + \bar{n}, \quad \bar{\Lambda}_c \rightarrow K_s^0 + \pi^0 + \bar{n}$$

Furthermore, the anti-deuteron $\bar{D}$ hadronization is commonly described by a coalescence mechanism [14], where a $\bar{p}$ and a $\bar{n}$ sufficiently close in phase space bind to form a $\bar{D}$. This latter is expected among the products of dark matter annihilation [15]; thus, the simultaneous measurement of the $\bar{n}$ - $\bar{p}$ correlation

function in $\bar{D}$ production would significantly improve the understanding of this subject, allowing a more complete analysis of dark matter distribution.

A deeper knowledge on anti-neutrons kinematic and interaction would improve also Monte Carlo GEANT4 models of the hadronization process. GEANT4 is the main toolkit designed for simulating the passage of particles through matter [16]. It finds applications across high-energy, nuclear and accelerator physics, as well as in medical and space science research.

1.2.1 Anti-neutron interactions in physics

Due to isospin symmetry, it was believed that anti-neutrons strong interaction with nucleons (N) and matter would not provide much more information than those obtained by anti-protons, for which obtaining proper beams was much easier.

The experimental landscape underwent a significant transformation in the mid-1980s with the development of dedicated anti-neutron beams, which enabled the observation of unique $\bar{n}$ characteristics, such as:

- The anti-nucleon nucleon $\bar{N}N$ interaction, in $\bar{p}p$ interactions, is a mixture of isospin $I = 0$ and $I = 1$ amplitudes, while the $\bar{n}p$ is a pure $I = 1$ eigenstate.
The same difference between $\bar{p}p$ and $\bar{p}n$ interactions could be obtained by means of the charge-symmetric system $\bar{p}n$ copiously produced at deuterium targets experiment. However, the use of deuterium introduces several drawbacks, such as re-scattering effects that may largely distort the final spectra [17]. This effect is absent if the $\bar{N}N$ system with pure $I = 1$ is formed by $\bar{n}$ interactions on protons of the material.
- The $\bar{n}p$ is a pure strong interaction, since it is free from Coulomb corrections, that are necessarily present in the $\bar{p}p$ interaction and may introduce further complications and ambiguities in the analysis of the data, especially at low-momenta.
- At low momentum, typically below 1 GeV/c, only S- and P-waves are present in the elementary $\bar{n}p$ interaction, described by the Dover-Richard parametrization [18]. Fig. 1.1 shows the trends of S-wave and P-wave annihilation cross sections according to the Dover-Richard parametrization, where in $\bar{n}p$ interactions the S-wave contribution diminishes with increasing energy, whereas the P-wave contribution grows.

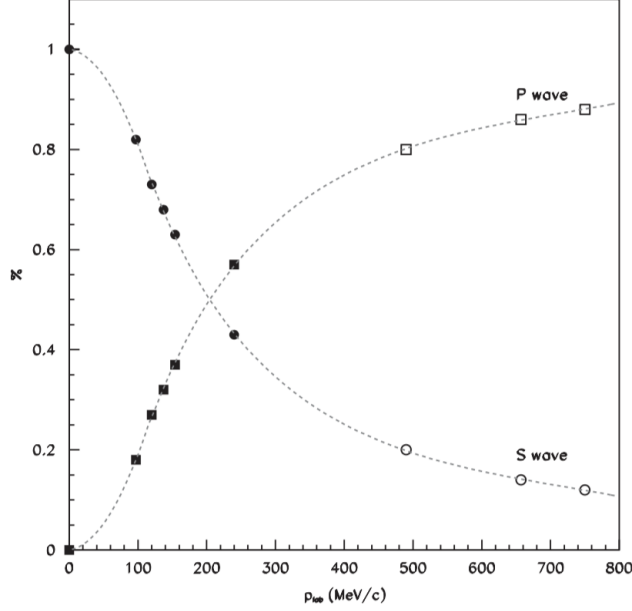


Figure 1.1: Trends of S- and P-wave annihilation cross sections according to annihilation models: the full markers (circles for S-wave, squares for P-wave) are from the Dover–Richard model [1], while the open ones have been obtained by [2] using a version of the optical model.

Therefore, the $\bar{n}p$ system can be represented as an isospin eigenstate $|I, I_3\rangle = |1, +1\rangle$. It is not a C-parity eigenstate, then only J (total angular momentum), P (spatial parity) and G-parity are good quantum numbers, where G-parity is defined as the combination of charge conjugation and a rotation of 180° in isospin space as $G = C e^{i\pi I_2}$.

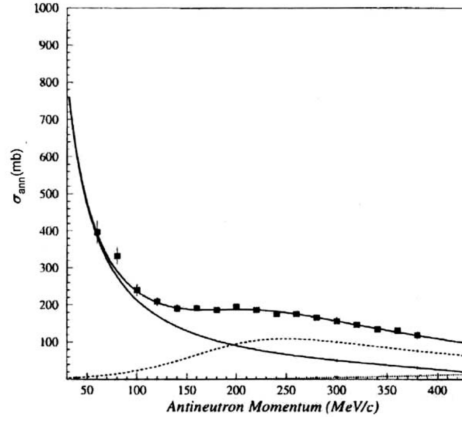
Since $\bar{n}p$ is an antifermion-fermion system, the following equations can be obtained:

$$P = (-1)^{L+1} \quad G = (-1)^{L+S+1}$$

where L is the relative angular momentum between the anti-neutron and the proton, while S is the spin of the system, namely 0 or 1. The $I^G(J^P)$ quantum numbers allowed for the lowest initial states are shown in table 1.

Even if the total cross section σ_T is the first piece of information on the interaction between two particles, for the case $\sigma_T(\bar{n}p)$ could be tough to measure it. The final value of σ_T published in [19] shows how the data could be fitted to a simple parametrization [20], obtaining the results showed in 1.2.

Wave	$2S+1L_J$	$I^G(J^P)$
S	1S_0	$1^-(0^-)$
	3S_1	$1^+(1^-)$
P	1P_1	$1^+(1^+)$
	3P_0	$1^-(0^+)$
	3P_1	$1^-(1^+)$
	3P_2	$1^-(2^+)$
D	1D_2	$1^-(2^-)$
	3D_1	$1^+(1^-)$
	3D_2	$1^+(2^-)$
	3D_3	$1^+(3^-)$

 Table 1: Quantum number $I^G(J^P)$ on S , P e D waves.

 Figure 1.2: Fit of $\sigma_{ann}(\bar{n}p)$ measured by OBELIX up to 400 MeV, including S- (solid line), P- (broken line) and D- (point line) wave contribution.

1.2.2 The anti-nucleon–nucleon interaction

The first approach to describing the antinucleon–nucleon ($\bar{N}N$) interaction relies on **interaction models**. Unlike Quantum Electrodynamics (QED), where the coupling constant α grows with energy and diverges at finite energy, asymptotic freedom characterizes the Quantum Chromodynamics (QCD), the quantum field theory of the strong force between quarks and gluons. Quarks interact weakly at high energies, enabling perturbative calculations, but at low energies the interaction strengthens dramatically, confining quarks and gluons within composite hadrons. Here, the divergent coupling constant α_s renders perturbative approaches invalid, making interaction models essential for describing strong interactions in the $\lesssim 1$ GeV energy range.

The core of this approach is that, following the $\bar{N}N$ annihilation, a *fireball* (which represents the $\bar{N}N$ system) with definite energy and volume $\Omega = \frac{4}{3}\pi R_0^3$

is formed, where $R_0 \sim 1$ fm is the nucleon radius. All the possible $\bar{N}N$ annihilation final states -primarily light mesons- may be reached through the decay of the fireball, according to the statistical Fermi model, where the probability of a final state is determined solely by the available phase space [21].

The use of interaction models in $\bar{N}N$ interaction generally allows a correct description of annihilation, even if it doesn't reproduce the whole experimental scenario, such as differences between nucleon and anti-nucleon scattering and stopping power (Barkas effect) [22]. The reason why it happens is that the potential formulation used to describe the short range interactions (where annihilation plays the most important role) is not sensitive to the details of the internal dynamics, namely the interactions between quarks and gluons.

The second approach is the **Effective Range** (ER) expansion one, which is independent on any hypothesis concerning the interaction potential and it is based on the obtained fit of experimental cross sections [23]. The annihilation cross section for two interacting particles may be expressed as a partial waves expansion as follows:

$$\sigma_{\text{ann}} = \sum_L \sigma_L^{\text{ann}} = \frac{4\pi}{k^2} \sum_L (2L+1) (\Im m f_L - |f_L|^2),$$

where k is the center-of-mass momentum of the incident particle, and the partial amplitude f_L may be written as a function of the δ_L wave phase shifts induced by the interaction in the system wave function:

$$f_L = \frac{1}{\cot \delta_L - i},$$

The partial waves may be expanded as a series in k , following the so-called *Effective Range (ER) expansion formalism*:

$$k^{2L+1} \cdot \cot \delta_L = \frac{1}{A} + \rho k^2 + O(k^4),$$

where A is the scattering length, and ρ is the interaction range. For S and P partial waves can be obtained:

$$k \cdot \cot \delta_0 = \frac{1}{a} + \frac{1}{2} r k^2$$

$$k^3 \cdot \cot \delta_1 = \frac{1}{b} - \frac{3}{2} \frac{1}{R} k^2$$

a and b are, respectively, the scattering length and nucleon volume, while r and R are the two effective ranges for the two partial waves. These parameters must be determined by a phenomenological fit to the annihilation cross section data. For example, the $I = 1$ source, relevant to the $\bar{n}p$ interaction case, it was found [20], respectively:

$$a_1 = (0.4 + 0.5i) \text{ fm and } r_1 = (-1.4 + 1.8i) \text{ fm for S-wave,}$$

$$b_1 = (0.8 + 0.1i) \text{ fm}^3 \text{ and } R_1 = (0.2 - 0.4i) \text{ fm for P-wave}$$

1.2.3 Electromagnetic and hadronic showers

In energy scenario beyond 10 MeV, Bremsstrahlung and pair production are the main processes which respectively involve photons and electrons/positrons, leading to the development of **electromagnetic showers** within the material. Therefore, to detect such phenomena, HEP experiments employ **electromagnetic calorimeters**. The electromagnetic calorimeter is a device whose size and density are designed to fully contain the electromagnetic shower, producing a signal proportional -or at least correlated- to the energy of the incident particle that initiated the shower. Since the radiation length X_0 determines the size of the electromagnetic shower, the typical longitudinal size for these detectors is of the order of $10\text{-}20 X_0$ ($\sim 30 - 40\text{cm}$).

Furthermore, **hadronic showers** can be originate from strong interactions of high energy (beyond $\sim GeV$) hadrons, such as protons, neutrons and pions, with the material, as well. To detect this kind of shower, an **hadronic calorimeter** may be employed; since the interaction length λ determines the hadronic shower size and $\lambda \gg X_0$, hadronic showers are larger and longer than electromagnetic ones, requiring hadronic calorimeters to be significantly bigger than electromagnetic calorimeters. Therefore, homogeneous calorimeters cannot be used for hadronic calorimetry as they are for electromagnetic calorimetry. Instead, the sampling calorimeter technique is mandatory, employing alternating layers of passive, dense absorber material and active detection material.

For example, approximately 95% of a hadronic shower is contained within a cylinder of radius equal to the hadronic interaction length ($\lambda_{had} \simeq 44.1\text{ cm}$, in CsI(Tl)). In contrast, about 90% of an electromagnetic shower is contained within a cylinder defined by the Molière radius ($R_M \simeq 3.6\text{ cm}$ in CsI(Tl)). Therefore, there is an order of magnitude difference between the two values. A shape comparison between these two type of shower can be observed in 1.3.

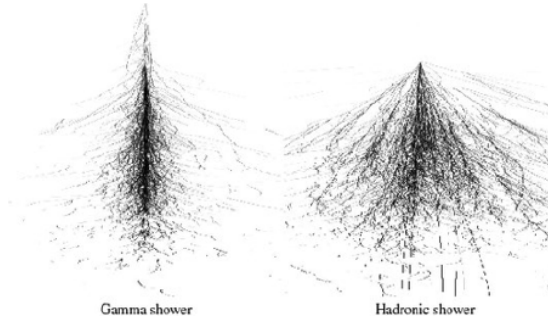


Figure 1.3: Electromagnetic shower vs hadronic shower comparison.

The $\bar{n}$ primarily interacts via the strong nuclear force, annihilating with nucleons in the material ($\sigma \sim 100\text{ mb}$), mainly producing light mesons such as pions which subsequently interact within the calorimeter:

$$\bar{n} + p/n \rightarrow \pi^\pm, \pi^0..$$

The π^0 mesons mainly ($\sim 98.8\%$) decays into a couple $\gamma\gamma$, producing electromagnetic showers that are fully contained in the calorimeter.

While charged pions π^+ and π^- undergo both electromagnetic and hadronic interactions, producing final products that are not fully contained within the calorimeter volume, this leads to a lack in the energy deposited in the detector. Both forward and backward directions are then involved, according to the reaction kinematics.

1.3 The MANTRA Project

It may be challenging for general-purpose experiments to directly identify and reconstruct anti-neutrons produced in e^+e^- collisions, such as those at Belle II and BESIII, which lack a hadronic calorimeter. Anti-neutrons generally annihilate with detector material nucleons such as that of an electromagnetic calorimeter, producing hadronic showers which only allows a loose measurement of the anti-neutron energy with extremely large uncertainty.

Since anti-neutron kinematics have so far been measured via missing mass methods or isospin symmetry with anti-protons, a direct anti-neutron reconstruction would be highly desirable, especially in studying processes with neutral hadrons, such as $B^+ \rightarrow K^+ n \bar{n}$ or $J/\Psi \rightarrow p \pi^- \bar{n}$.

This thesis fits in the larger context of the **MANTRA** (Measuring Anti-Neutron: Tagging and Reconstruction Algorithm) PRIN2022 project, an inter-university (University of Turin, University of Ferrara and INFN) effort aimed at developing a method to improve the reconstruction of anti-neutron in existing high-energy physics experiments. It provides a general approach to measure the energy of **anti-neutrons** $\bar{n}$ up to 10 GeV using an electromagnetic calorimeter.

The main idea is to combine the information from a high time-resolution time of flight detector (TOP and TOF in the Belle II and in the BESIII experiments, respectively) with that from an electromagnetic calorimeter (ECL and EMC in the Belle II and in the BESIII experiments, respectively), where the anti-neutrons annihilation occurs.

Due to its neutral charge and its immediate annihilation, tagging $\bar{n}$ can be very challenging. However, the annihilation can produce self-contained light mesons, mostly made of high-energy pions (π^-, π^0, π^+) and soft nuclear fragments. These can be detected in the Electromagnetic Calorimeter (detailed examination on next chapter), where charged pions can produce hadronic showers, while neutral pions immediately decay into two photons which produce electromagnetic showers in the calorimeter volume. The comparison of the cluster energy deposition patterns, the so called shower shapes variables, left in the ECL by anti-neutrons and other neutral particles, such as photons, can improve the characterization of neutral particles.

A crucial aspect of the analysis process is the role of simulation, where GEANT4 [16] (more details in the next chapter) plays a central role, modeling the hadronic interactions in the calorimeter; realistic simulation is essential to accurately reproduce real data. For $\bar{n}$ annihilation, parameterizations are typically derived from $\bar{p}$ measurements, necessitating validation against data.

This thesis aims to study the $\bar{n}$ hadronic interactions modeled by simulation software and compare them with those observed in Belle II data, focusing exclusively on signals from the Belle II electromagnetic calorimeter. To achieve this, $\bar{n}$ variables are first studied using a signal sample, where only the specifically selected processes with $\bar{n}$ in the final state are generated. Several background channels are then included to simulate the complete Belle II environment, concluding with a comparison between the results obtained from simulations and those from real data, with a Data/MC agreement.

Therefore, this work addresses the following question:

Are $\bar{n}$ hadronic showers correctly simulated in the Belle II software?

2 The Belle II experiment

The following sections introduce the **SuperKEKB** [24] collider, describing its technical setup briefly. It then presents the **Belle II** [5] detector discussing it sub-detector by sub-detector. Afterwards, the Belle II Analysis Software Framework (**basf2**) and **TopoAna** program are briefly described, with attention to the role of Monte Carlo methods in collider experiment simulation and reconstruction.

2.1 The SuperKEKB collider

SuperKEKB [24] is an asymmetric e^+e^- double-ring collider, located at KEK (The High Energy Accelerator Research Organization), in Tsukuba, Japan. It is the upgrade of the previous collider KEKB. It has a circumference of 3016 m and collides bunches of electrons with bunches of positrons, respectively of energy 7 GeV (High Energy Ring -HER-) and 4 GeV (Low Energy Ring -LER-). Fig. 2.1 is a schematic image of the SuperKEKB design:

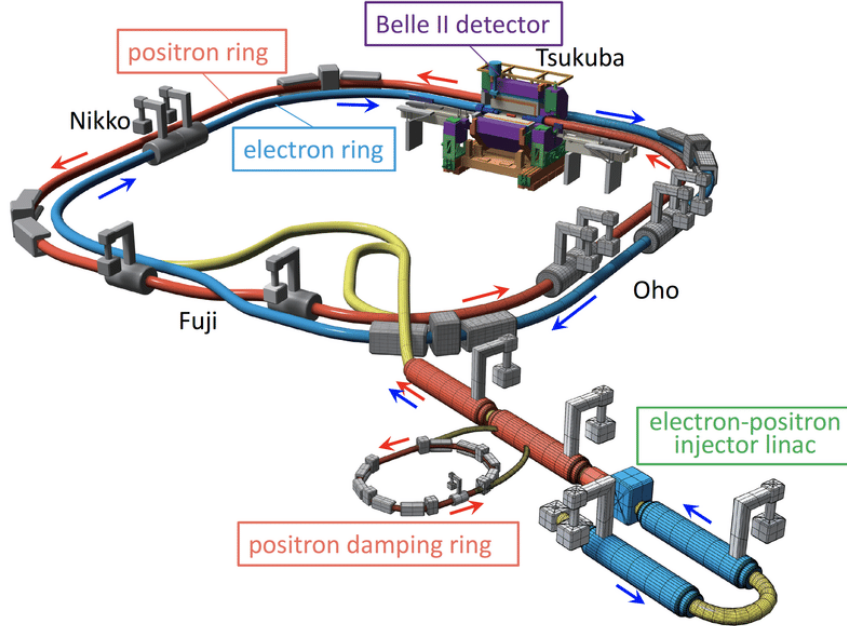


Figure 2.1: SuperKEKB design, where the Interaction Point (IP) is located in Tsukuba surrounded by the Belle II detector.

It is noteworthy that the positron damping ring reduces the e^+ beam emittance, as positrons are obtained from a rotating tungsten target subjected. Electrons, by contrast, are injected directly from the linac since a controlled e^- beam can easily be obtained [25].

The resulting center-of-mass energy $\sqrt{s}$ in the IP can be modulated, but most of the data are acquired at the $\Upsilon(4S)$ resonance (10.58 GeV), an excited quarkonium state of $b\bar{b}$, which mainly decays into an entangled couple of $B\bar{B}$ ($\sim 100\%$). The $\Upsilon(4S)$ mass is just 20 MeV above the $B\bar{B}$ threshold, then the mesons are produced nearly in the rest frame. Since the lifetime of neutral B-mesons is short ($\tau = 1.510$ ps) and they are produced with low velocities, their path before decaying is short, making tough to separate the decays of the two produced mesons. The explicit asymmetry of bunches results in a relativistic forward Lorentz boost of the center-of-mass, where the decay length of the mesons is longer in the LAB frame, allowing the observation of the produced B mesons. Due to the huge amount of produced B mesons, SuperKEKB is regarded as an example of so-called (super) **B-factory**.

SuperKEKB has been running since 2018 and it achieved the world's highest instantaneous luminosity ($5.244 \times 10^{34} \text{ cm}^{-2} \text{ s}^{-1}$) in March 19, 2026. The design luminosity of SuperKEKB $8 \times 10^{35} \text{ cm}^{-2} \text{ s}^{-1}$, 40 times higher than Belle's one, with the goal to accumulate data corresponding to an integrated luminosity of 50 ab^{-1} . The March 19, 2026 SuperKEKB 24-Hour Operation Summary is shown in Fig. 2.2.

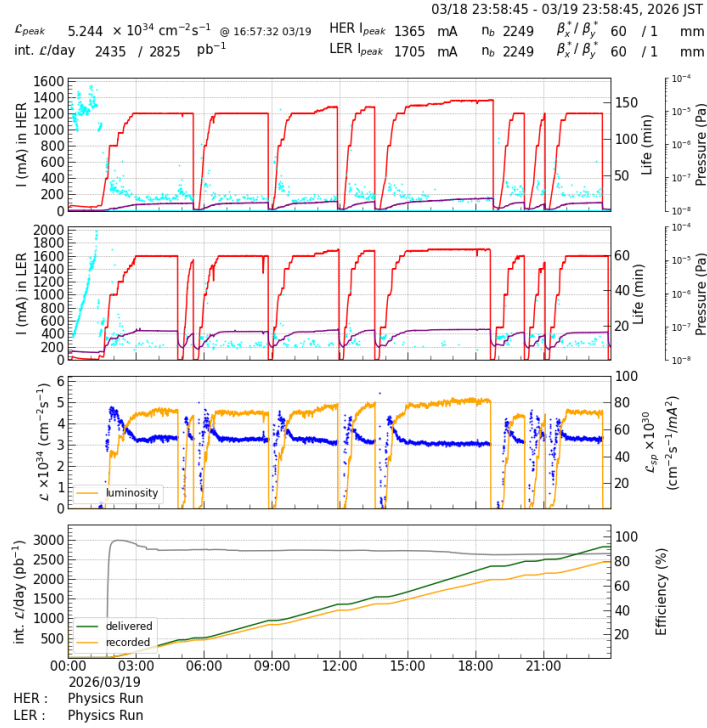


Figure 2.2: SuperKEKB operating parameters on March 18-19, 2026. From top to bottom: HER and LER ring currents (first two panels), instantaneous and integrated luminosity (bottom two panels)

2.2 The Belle II detector

The only interaction point of SuperKEKB is surrounded by the **Belle II** detector [5], a general purpose high-energy particle layered detector with almost 4π of solid angle coverage. It presents a cylindrical shape to cover the e^+e^- collisions, composed of a barrel and two end-caps located along the z-axis.

The Belle II detector measures 8 m in both width and height, with a total weight of approximately 1400 tons. With the exception of the KLM, all its subsystems are immersed in a 1.5 T magnetic field.

The detector is composed of several sub-detectors arranged cylindrically, reflecting the asymmetric distribution of the final state particles. The cylindrical structure of the detector is the *barrel*, while the bases are made of the forward end-cap (along the direction of the e^- beam momentum) and the backward end-cap (along the direction of the e^+ beam momentum).

The Belle II detector consists of the following sub-detectors, arranged radially outward from the IP:

- (a) Beam Pipe made of Beryllium (Be),
- (b) Two layers of silicon pixel detector (PXD),
- (c) Four layers of double-sided silicon strip detector (SVD),
- (d) The Central Drift Chamber for tracking the charged particle (CDC),
- (e) Time of propagation (TOP) Cherenkov counter in the barrel + Cherenkov ring imaging (ARICH) in the front end-cap region,
- (f) The electromagnetic calorimeter (ECL) to detect photons and to identify charged particles,
- (g) Superconductive magnet which provides 1.5 T axial magnetic field in order to bend charged particle,
- (h) K_L and muon detector (KLM) to track muons and K_L .

The electromagnetic calorimeter (ECL) has a central role in this elaborate since neutral ECL clusters associated to anti-neutron candidates are studied. A Belle II scheme is shown in Fig. 2.3.

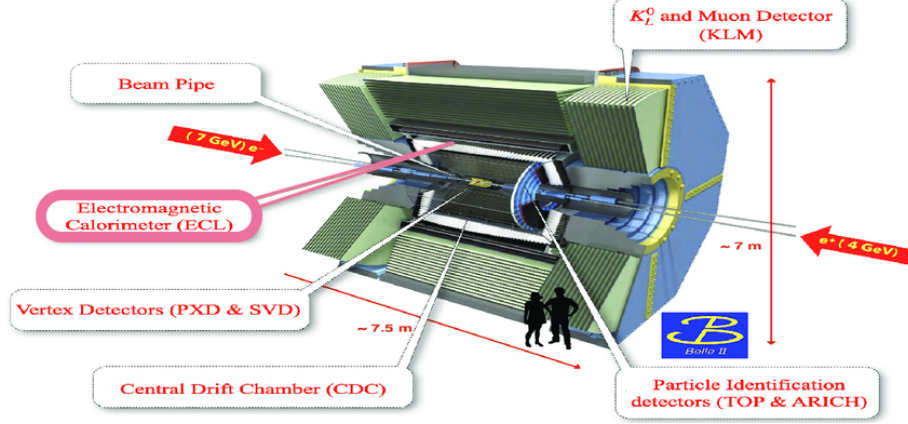


Figure 2.3: An artistic view of the Belle II detector

2.2.1 The vertex detector VXD

Immediately following the beryllium beam pipe, which does not play a detection role, the innermost layer is the Vertex Detector (**VXD**).

It consists of two devices: the Silicon Pixel Detector (**PXD** - 2 layers) [26] and the Silicon Vertex Detector (**SVD** - 4 layers) [27]. Overall there are six layers, of whose the first two ($r_1 = 14\text{mm}$, $r_2 = 22\text{mm}$) are pixelated sensors of the DEPFET type and the remaining four ($r_3 = 38\text{mm}$, $r_4 = 88\text{mm}$, $r_5 = 115\text{mm}$, $r_6 = 140\text{mm}$) are double-sided silicon strip sensors.

The VXD provides coverage of $17^\circ < \theta < 150^\circ$ and full azimuthal coverage in ϕ , where θ is the polar angle relative to the z -axis -aligned with the e^- beam direction- and ϕ is the azimuthal angle in the xy plane using cylindrical coordinates. A scheme of the VXD is shown in Fig. 2.4.

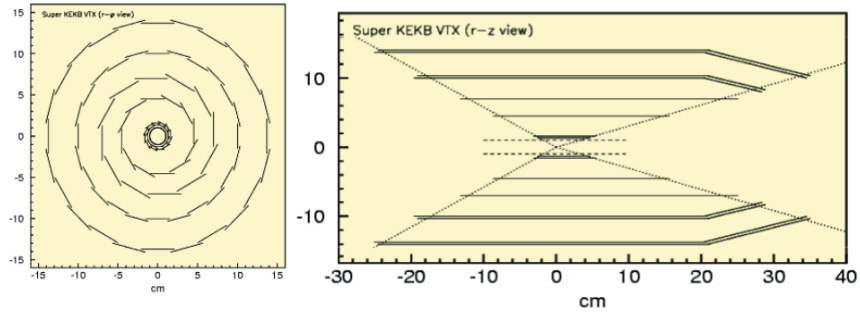


Figure 2.4: A scheme of the Belle II VXD

The VXD purpose is to reconstruct the position of the primary vertex, the vertex of B-mesons decay which are directly produced by the decay of the $\Upsilon(4S)$, improving the resolution of the tracking system as well.

2.2.2 The Central Drift Chamber (CDC)

The Central Drift Chamber (**CDC**) [28] is a large volume composed of small drift cells and it is the main tracking system of the Belle II detector. The CDC reconstructs charged-particle tracks by measuring the hit positions, allowing the determination of their momenta. Furthermore, the specific energy loss dE/dx measurement provides PID (Particle IDentification) information, complemented by the 1.5 T magnetic field that enables charge separation.

The CDC has an internal radius of 16 cm and an external radius of 113 cm and it contains 14339 wires in total distributed in 56 layers, either parallel (axial orientation) or skewed (stereo orientation) to the magnetic field. Combining their information, it is possible to reconstruct a full tridimensional helix track. Inside the cell, the average drift velocity is $3.3\text{cm}/\mu\text{s}$.

The CDC has the same VXD angular (θ, ϕ) coverage and, in order to increase the acceptance at forward angles, it is implemented asymmetrically along the z axis. It is filled with a mixture of helium (He) and ethane (C_2H_6), which represents a proper trade-off between momentum resolution for tracking and dE/dx resolution for the PID. A schematic view of the CDC is shown in Fig. 2.5:

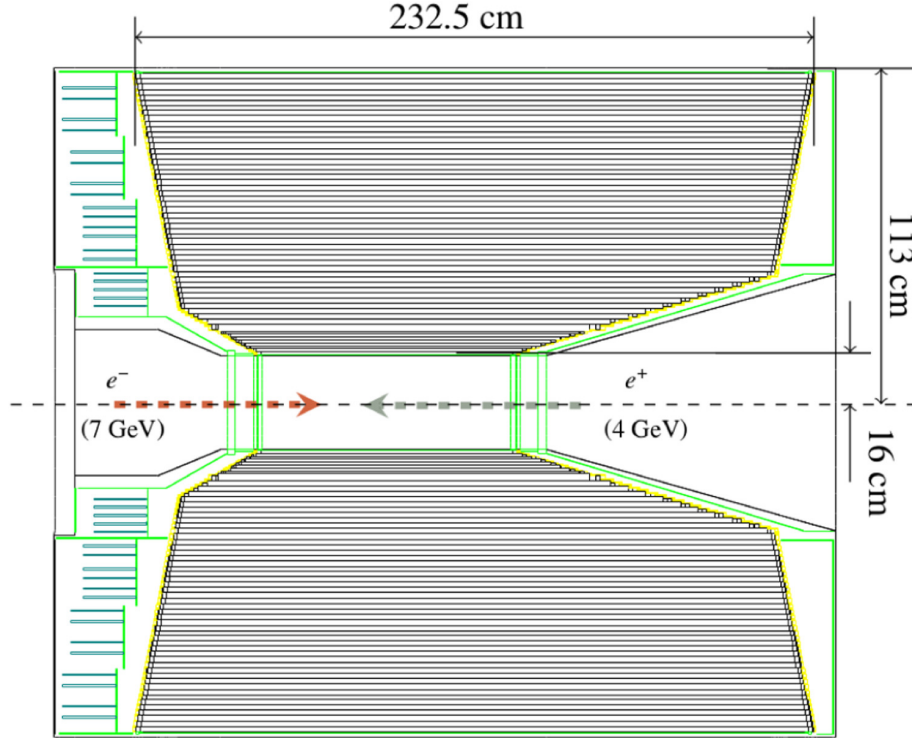


Figure 2.5: A scheme of the Belle II CDC.

2.2.3 PID systems: TOP and ARICH

Particles crossing a Cherenkov radiator material at speeds exceeding the phase velocity of light in that medium - $\beta > 1/n$, where n is the refractive index- emit photons at the Cherenkov angle $\theta_c = \arccos(1/\beta n)$. Particles with identical momentum but different masses produce Cherenkov radiation at distinct angles, enabling particle identification (PID).

For particle identification in the barrel region, a Time of Propagation (**TOP**) [29] counter -a novel Cherenkov detector- is employed. In total, there are 16 modules, each consisting of a 45 cm wide and 2 cm thick quartz bar with a small expansion volume of about 10 cm long at the sensor end. There is an expansion wedge that behaves as a lens and improves the spatial resolution of the Cherenkov photon imaging. At the exit window of the wedge, two rows of sixteen fast multi-anode photon detectors are mounted.

Within the 2.6 m-long quartz, the particle produces Cherenkov radiation that travels via total internal reflection to the mirror. Then the radiation is reflected towards the photon detectors at the backward end, where a small prism is located, acting as a volume expansion and a light guide. The time of arrival and the impact position of the Cherenkov photons provide the PID information.

The hit distributions differ according to the particle species and their masses: different particles exhibit distinct Cherenkov angles, resulting in different photon paths. Consequently, they are detected at different positions within the PhotoMultiplier (PMT) array and at different times. A scheme of TOP counter module is illustrated in Fig. 2.6 and a side view of the TOP counter in 2.7.

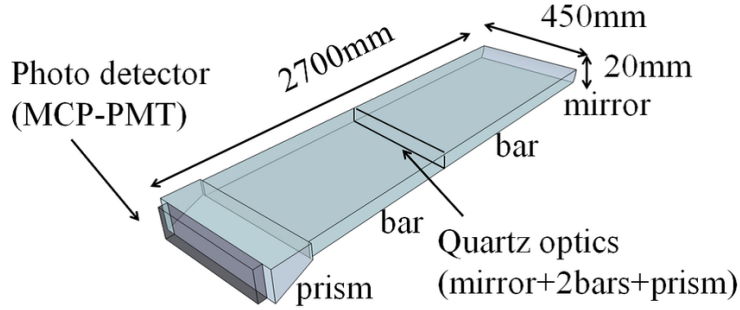


Figure 2.6: Scheme of the TOP counter module.

Thanks to a 16-channel Micro Channel Plate PhotoMultiplier (MCP-PMT), TOP counters are equipped with photosensors that provide a time resolution of about 100 ps.

The Aerogel Ring Imaging Cherenkov (**ARICH**) [30] is located in the forward end-cap region of the Belle II detector and is thus complementary to the aforementioned TOP. Its main goal is to separate kaons K from pions π across the full kinematic range of the experiment, from 0.5 to 4.0 GeV/c .

Since particles with the same momentum but different mass radiate at different angles, it is designed to detect and distinguish charged particles.

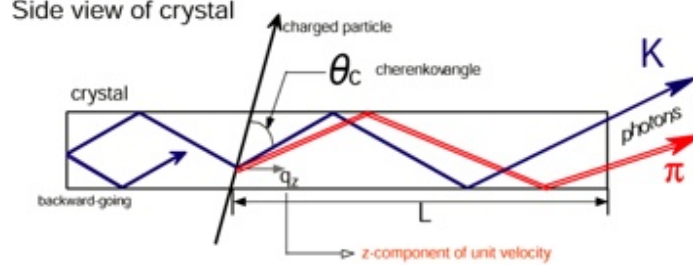


Figure 2.7: Representation of the side view of the TOP counter with internal reflection of the Cherenkov radiation.

It consists of two 2 cm thick aerogel layers ($n_{up} = 1.045$, $n_{down} = 1.055$), which increase the photon yield without degrading the Cherenkov angle resolution compared to using a single aerogel type [31]. A scheme of the ARICH working principle and an example of a ring produced by a cosmic-ray muon are shown in Fig. 2.8.

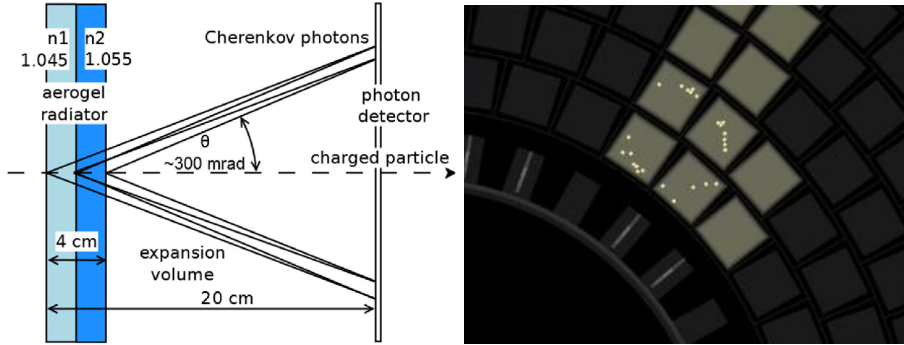


Figure 2.8: (a) Working principle of the ARICH. (b) Ring produced by a cosmic-ray muon.

2.2.4 The Electromagnetic Calorimeter (ECL)

The Electromagnetic CaLorimeter (**ECL**) [32] plays a central role in this thesis, since the neutral ECL clusters associated to anti-neutron candidates are studied. A typical shape of an ECL shower caused by a 7 GeV photon is shown in Fig. 2.9. The primary purpose of the ECL is to detect photons and electrons, providing precise measurements of their energy and position (E , $\vec{x}$), studying the produced electromagnetic shower. The ECL contributes to identifying neutral particles -such as photons and neutral hadrons. as well as separate the charged particles, primarily electrons and pions. Anti-neutrons can be detected through its annihilation products (mainly pions).

The Molière radius R_M is a characteristic constant of a material that defines

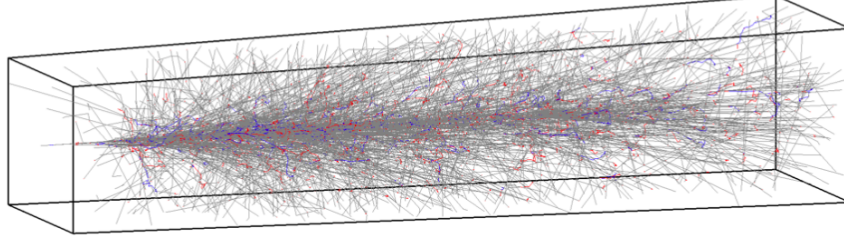


Figure 2.9: Typical ECL shower caused by a 7 GeV photon.

the transverse scale of fully contained electromagnetic showers initiated by high-energy electrons or photons. Its definition, for an electromagnetic calorimeter, is that $\sim 90\%$ of electromagnetic shower is contained in a cylinder with radius R_m , while the $\sim 95\%$ is contained in a cylinder with radius $2R_m$.

The radiation length X_0 is defined as the mean length (in cm) over which an electron's energy is reduced by a factor of $1/e$. The critical energy E_c is defined as the energy at which radiation losses equal ionization losses for electrons/photons. X_0 and E_c determine the Molière radius R_M as follows:

$$R_m = \frac{21MeV}{E_c} \times X_0$$

CsI(Tl) crystals

The thallium-doped caesium iodide **CsI(Tl)** crystals have $X_0 = 1.860$ cm and $R_m = 3.53$ cm. This material is one of the brightest known scintillator, providing a light yield Y of ~ 52000 photons/MeV, with a photon emission spectrum at ~ 550 nm, ideal for photodiode readout.

Material	Density (g/cc)	Wavelength of Maximum Emission	Refractive Index	Principal Decay Constant (μ -Sec)	Total Light Yield	Pulse Rise Time (μ -Sec)	Absolute Scintillation Efficiency	Relative Pulse Height
NaI(Tl)	3.67	415	1.85	0.23	38000	0.5	11.3	1.00
CsI(Tl)	4.51	540	1.80	1.0	52000	4.0	11.9	0.49
LiI(Eu)	4.08	470	1.96	1.4	11000		2.8	0.23
BGO	7.13	505	2.15	0.30	8200	0.8	2.1	0.13
BC-400	1.032	423	1.58	0.002	10000		3.0	0.25
BC-454	1.026	425	1.58	0.0022	10000		3.0	0.25

Figure 2.10: Properties of inorganic scintillators used in calorimetry [3].

The ECL is composed of 8736 array of **CsI(Tl)** crystals. Each is a truncated pyramid with dimension $6 \times 6 \times 30$ cm^3 , where the radial length is around $16.2X_0$.

Each crystal is wrapped with a layer of $200\ \mu\text{m}$ thick Gore-Tex porous teflon and then covered with a $50\ \mu\text{m}$ thick aluminized polyethylene.

Assuming the z -axis is aligned with the e^- beam momentum direction, with $z = 0$ coinciding with the nominal IP position, the ECL is made of a 3 m long barrel section and the end-caps at $z = 2.0$ m (forward) and $z = -1.0$ m (backward). An ECL scheme is shown in Fig. 2.11:

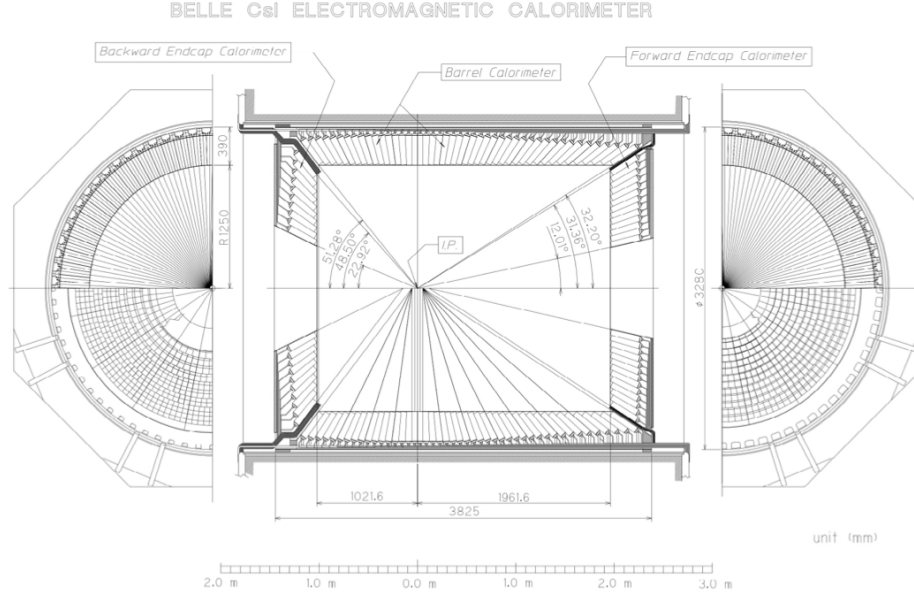


Figure 2.11: A scheme of the Belle II ECL detector.

It covers about 90% of the solid angle in the center-of-mass system (polar coverage: $12^\circ \leq \theta \leq 155^\circ$); there are a $\sim 1^\circ$ gap between barrel and each endcap.

An image showing the ECL θ coverage and segmentation for each region is shown in Fig. 2.12.

	θ coverage	θ segments	ϕ segments	# of counters
Forward Endcap	$12.01^\circ - 31.36^\circ$	13	48 - 144	1152
Barrel	$32.2^\circ - 128.7^\circ$	46	144	6624
Backward Endcap	$131.5^\circ - 155.03^\circ$	10	64 - 144	960

Figure 2.12: ECL coverage and segmentation for each ECL region.

2.2.5 The solenoid and iron structure

Following the ECL, an axial magnetic field of 1.5T is generated by a 23 tons superconducting solenoid. The curvature of charged-particle trajectories (Larmor radius) allows the determination of their momentum and charge, which, when combined with additional detector information, enables particle identification. The magnet is made of niobium-titanium-copper (NbTi/Cu) alloy and is cooled by a liquid helium cryostat to achieve superconducting conditions.

2.2.6 The K_L and Muon Detector (KLM)

The K_L and Muon Detector (**KLM**) [33] represents the outermost layer of Belle II and it is dedicated to the detection of K_L mesons and muons, which are very penetrating particles. It is composed of alternating layers of 4.7 cm thick iron plates and active detector elements; the former acts as a return yoke for the magnetic flux. The alternating layers provide at least 3.9 interaction lengths (compared to 0.8 in the ECL), allowing the K_L to interact and produce hadronic showers.

KLM covers the barrel region (15 layers) and the end-caps regions (forward 14 layers and backward 12 layers). The two inner barrel layers and the entire end-caps are instrumented with scintillator strips, while the 13 outer barrel layers are instrumented with Resistive Plate Chambers (RPC).

As compared to charged hadrons, muons travel longer distances and with smaller deflections inside the KLM, which allows discrimination between the two.

3 The Belle II Analysis Software Framework



To perform the simulations, reconstruction and the analyses of this thesis, the Belle II Analysis Software Framework (**basf2**) [34] has been employed. It is the primary software used in all aspects of the data-processing chain at Belle II, such as generating simulated data, unpacking raw data, performing reconstruction (tracking, clustering) and carrying out high-level analysis. The final analysis steps (as histogramming, fitting distributions, etc.) are usually performed in other frameworks as well, such as **ROOT** or the Matplotlib Python library.

3.1 Monte Carlo simulation

The Monte Carlo method is a technique used to compute probabilities and related quantities by exploiting sequences of pseudo-random numbers. Sequences of random numbers possess meaning, not individual ones—though the latter are frequently referenced. Randomness characterizes the sequence, not single numbers.

Monte Carlo methods are widely used in HEP physics for tasks such as **event generation** and **detector simulation**. The first one regards the production and the decays of particles with a certain set of properties, while the second one concerns the simulation of the particle interactions during the path in the detector materials.

In Belle II, electron-positron collisions produce particles according to the adopted physics model, which may follow the Standard Model or alternatives such as supersymmetry or dark matter production. Example of a physics generator could be *PYTHIA* for the generation of high-energy physics collision events, *EvtGen* that simulates the decays of heavy flavour particles, primarily B and D mesons and *Phokhara* which simulates the leptonic and hadronic final states at the next-to-leading order (NLO) accuracy, namely simulating ISR radiation, which will have a primary role in the next chapters, and so forth.

After particle generation, the simulation of propagation through the detector must be addressed, including interactions with active materials and the material budget. This is powered *GEANT4*, an open source software which performs the interaction between particles and the detector systems.

3.2 Reconstruction

The reconstruction process allows the particles measurement and identification. The closer the reconstructed quantities (such as initial positions and four-momenta) are to the generated-level information, the better the reconstruction performance in simulated MC and real data. Since several products of e^+e^- are short living particles, it is necessary to reconstruct them indirectly via their daughter particles. Furthermore, intrinsic noise must be taken into account, as it may generate spurious random signals unrelated to actual particles. Interactions between beam electrons or positrons, either with each other or with other accelerator components -known as beam background- must be considered.

The same reconstruction algorithms are employed in data and MC, in order to obtain the same reconstruction: in order to test the correct response of the reconstruction, the so called **MCTruth** are use used, which basically is the generator-level Monte Carlo truth of the simulated data pool. The Monte Carlo matching (MC-matching) procedure identifies the generator-level particle responsible for a reconstructed candidate, crucial for understanding reconstruction effects and backgrounds. This corresponds to the association of reconstructed particles with MC particles in Belle II nomenclature.

3.2.1 The ECL reconstruction algorithm

as mentioned, the main tasks of an electromagnetic calorimeter are to measure the energy and position of electromagnetic showers and to identify neutral and charged particles. For each crystals, a 32 bit word is the result of the signal, namely an **ECLDigit**. It is made of the energy information in GeV (18 bit), the time information in ns (12 bit) and 2 status bits (quality of the signal). In order to reconstruct the shower, there is an algorithm which converts the ECLDigit into ECLShowers and the into ECLClusters.

The first step is to identify the **connected regions**, which allows a rapid reduction of the combinatorics. An example of this procedure is shown below:

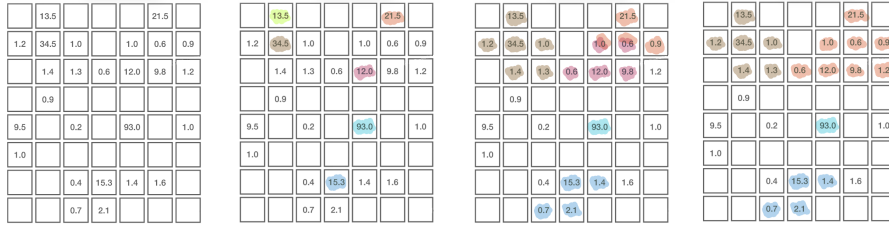
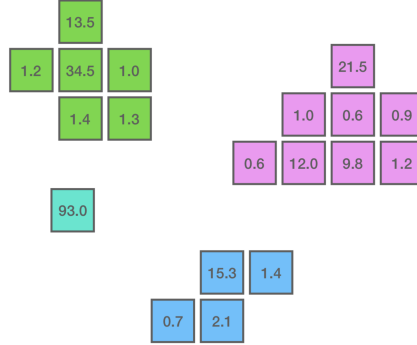


Figure 3.1: Description of ECL clustering process:(a) Step 0 (b) Step 1 (c) Step 2 (d) Step 3

Each cell represents one ECLDigit, whose value corresponds to the energy in MeV. Empty cells are below the readout threshold (0.1 MeV for this example).

In step 1, every ECLDigit above 10 MeV is selected. In step 2, any neighbors above 0.5 MeV are attached to the highest-energy seed found in step 1. Sometimes two seeds compete for the same neighbor, and sometimes clusters have no neighbors (i.e. 93.0). In step 3, overlapping neighbor regions trigger a merge. Additionally, any cells above 1.5 MeV recruit their next neighbors iteratively. After this process, the connected regions are obtained:



The final step is to pick the crystals with energy above 10 GeV, namely the Local Maxima (LM). Every connected region has one LM at least. The connected regions are furtherly processed in order to obtain the position and the energy of the cluster, processing the LM with a splitting-weighting process among the connected regions crystals [35].

The main idea is that an **ECLShower** object is produced from every LM. If there is only one local maximum within a connected region, the full energy of each crystal is associated with the corresponding ECLShower. Otherwise, if multiple local maxima are present, an iterative procedure is performed, where the weight for each crystal depends on the distance of the crystal from the center-of-gravity of each ECLShower (center-of-gravity is calculated using weighted energies).

Connected regions are used to minimize the number of times the computationally intensive splitting procedure must be performed.

3.2.2 ECL performance

The observed ECL **energy resolution** can be written as follow:

$$\sigma_E = \sqrt{\left(\frac{0.81\%}{\sqrt[3]{E}}\right)^2 + \left(\frac{0.066\%}{E}\right)^2 + (1.34\%)^2}$$

Where the first term represents stochastic fluctuations (Poissonian statistics) in shower generation, the second term accounts for electronic noise, and the third is a constant term dependent on calibration.

For example, $\frac{\sigma_E}{E} = 4\%$ at 100 MeV and $\frac{\sigma_E}{E} = 1.4\%$ at 8 GeV, while the angular

resolution is 13 mrad (3 mrad) at low (high) energies. For example, the π^0 mass is 135 MeV and its mass resolution is approximately 4.5 MeV, in absence of background (detailed description can be found in [36]).

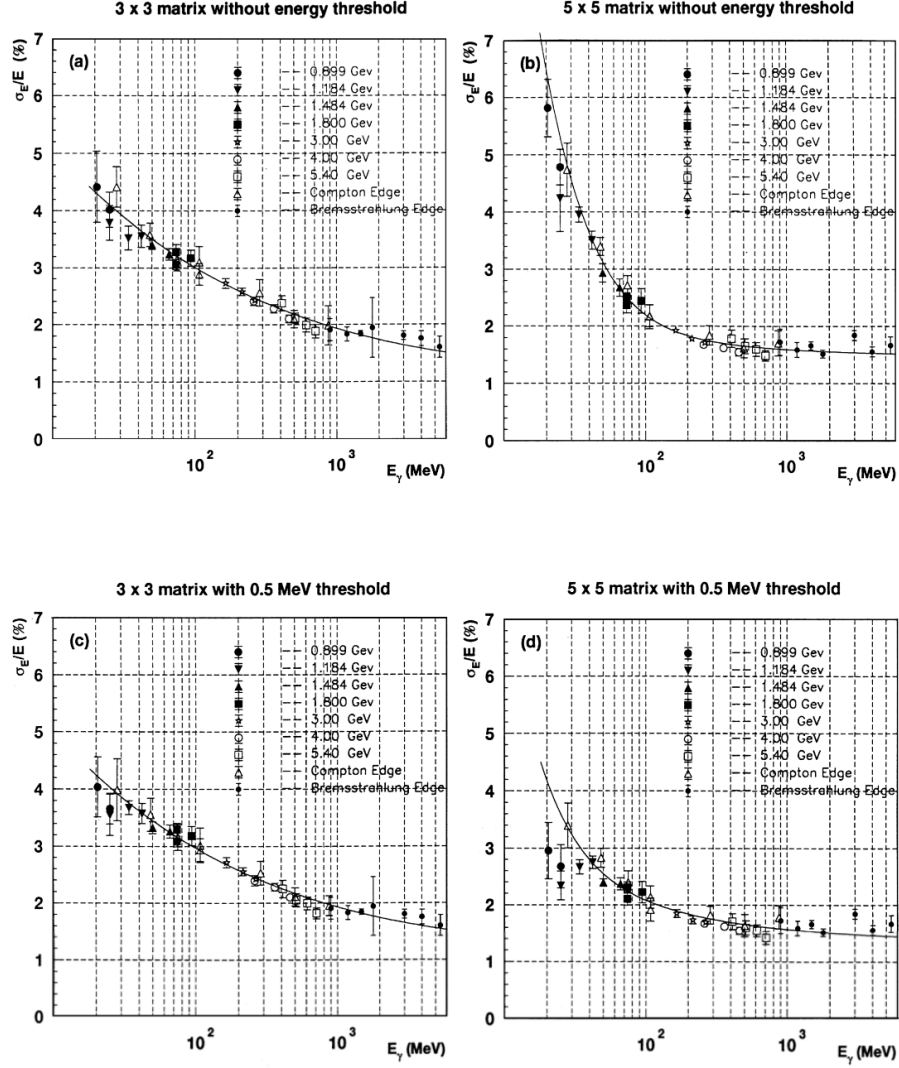


Figure 3.2: The energy resolution as a function of incident photon energy for the (a) 3Y3 and (b) 5Y5 matrix with total energy sum. The energy resolution of the (c) 3Y3 and (d) 5Y5 matrix with a 0.5 MeV threshold. The solid lines are the results of the χ^2 fit. The error bars are not statistical but are the rms of four measurements taken with the crystals (3,3), (3,4), (4,3) and (4,4) into the photon beam [4]

3.3 basf2 modules and path

A **basf2 module** is the smallest building block of the framework and it is a script, usually written in C++, designed to process a specific unit of data. A typical data chain consists of a linear arrangement of modules. All the work is done in modules, including reading and writing data to and from disk or more complex operations such as full detector simulation.

Modules are linearly arranged in a **path**, which is a container for them. The linear order depends on the specific task defined by the user. Therefore, the modules are executed one at a time, in the same order they were added into the path, from the first to the last. A schematic view of processing flow is shown below:

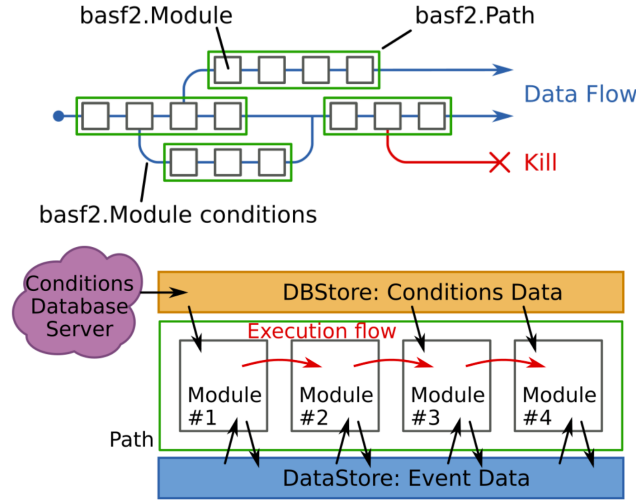


Figure 3.3: Scheme of the processing chain in basf2

The modules process data stored in the DataStore, with each module having access to read from and write to it. There is also a non-event data, the so called conditions, which is loaded from a central conditions database and is available in the DBStore. These databases stores the information about the specific detector condition at each run of the experiment.

A **package** is a logical collection of code in basf2, consisting of several modules and some python scripts that configure paths to do common things. The packages have meaningful name related to their specific operation (tracking, reconstruction, ecl, klm, etc...). All the packages are stored in the Belle II GitLab directory. The most used package in this thesis is the **analysis package**, which contains python functions and C++ basf2 modules, such as modularAnalysis. The **steering file** is the core script in which analysis and data-processing tasks are performed, with a path being declared and modules added to it. Some highlights of the steering files used are provided in the appendix.

3.4 TopoAna: Topology Analysis

In high-energy physics data analysis, a comprehensive understanding of inclusive MC samples could be crucial to select signals with higher efficiency while suppressing backgrounds. This is exactly the case of the present elaborate. Knowledge of the physics processes enables identification of the main backgrounds. Selection criteria can then be further optimized by analyzing differences between these backgrounds and the signal.

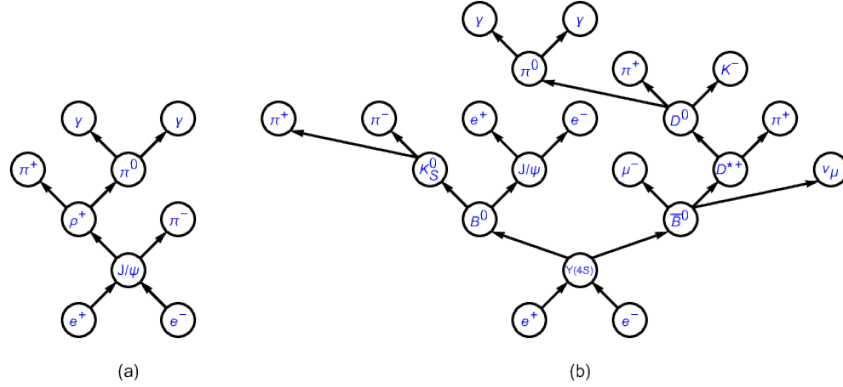


Figure 3.4: Topology diagrams of (a) $e^+e^- \rightarrow J/\psi$ and (b) $e^+e^- \rightarrow \Upsilon(4S) \rightarrow B^0 \bar{B}^0$, two typical processes at e^+e^- colliders, such as Belle II and BESIII

In order to have a very quick way to verify the Monte Carlo quickly and easily, a topology analysis program called **TopoAna** is developed with C++, ROOT, and LaTeX.

The program input consists of one or more ROOT files containing a **TTree** object with raw topology truth information from the inclusive MC samples under study. Specifically, each **TTree** entry includes three key ingredients for particles produced in an event: the particle multiplicity, their PDG codes, and their mother indices.

TopoAna is an offline tool independent of basf2. It independently extracts human-readable topology information from raw MC truth data via the MC-GenTopo interface. Referring to figure 3.4 (b):

4 Study of the signal sample channel

The first part of the present work aims to characterize the kinematics of the $\bar{n}$, using a Monte Carlo signal sample generated specifically for the selected reference channel:

$$e^+ + e^- \rightarrow p + \pi^- + \bar{n} (+ \gamma_{\text{ISR}})$$

This is selected because it features simple kinematics, with the $\bar{n}$ as the only missing particle, enabling a semi-exclusive analysis.

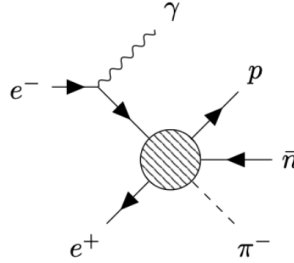


Figure 4.1: Feynman Diagram of the analyzed channel. The ISR gamma is a pre-interaction radiation emitted by the e^+ and/or e^- bunches. The recoil system is made of $p\pi^-(\gamma_{\text{ISR}})$

In particular, the kinematic relation between the $\bar{n}$ and the recoil system $p\pi^-(\gamma_{\text{ISR}})$ will be exploited, considering both configurations: in the presence of Initial State Radiation (ISR) photon emission and in its absence. The ISR is emitted by electrons and positrons in the initial bunches, which lose energy before the collision. This phenomenon induces a lowering of center-of-mass energy, which will no longer be the usual ~ 10.58 GeV. The ISR energy and angle distributions will be discussed in detail, providing a comparison with a simple phase space model emission.

Then the effect of a one constraint (1C) kinematic fit over the recoil mass will be discussed, in order to obtain an higher recoil momentum resolution.

100000 events are generated for the selected channel and to perform the event generation and the analysis the aforementioned **Belle II Analysis Software Framework (basf2)** is employed. To check the script code highlights related to this section and the adopted *phokhara_evtgen_combination* generator, please refer to Appendix A, where the decay file, the Monte Carlo generation file and the steering file are explained in detail.

4.1 Kinematic variables comparison

As a first step, the kinematic variables are studied through the recoil system consisting of $p\pi^-(\gamma_{\text{ISR}})$, where these particles are reconstructed. In parallel, the

neutral cluster associated to the $\bar{n}$ is reconstructed and its properties are compared to those from the recoil, comparing different kinematic variables. Through the recoil analysis, the antineutron is identified as the missing particle in the final state.

Before discussing the analysis results, an overview of the generated events is provided using the Belle II **TopoAna** software display:

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$vpho \rightarrow \pi^- \bar{n} p$	$\pi^- \bar{n} p$	0	35291	35291
2	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p$	$\pi^- \bar{n} p \gamma^I \gamma^I$	2	22971	58262
3	$e^+ e^- \rightarrow vpho \gamma^I, vpho \rightarrow \pi^- \bar{n} p$	$\pi^- \bar{n} p \gamma^I$	1	18735	76997
4	$vpho \rightarrow \pi^- \bar{n} p \gamma^F$	$\pi^- \bar{n} p \gamma^F$	3	10005	87002
5	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^I \gamma^F$	6	5274	92276
6	$e^+ e^- \rightarrow vpho \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^F$	4	4621	96897
7	$vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^F \gamma^F$	7	1503	98400
8	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^I \gamma^F \gamma^F$	8	700	99100
9	$e^+ e^- \rightarrow vpho \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^F \gamma^F$	5	597	99697
10	$vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^F \gamma^F \gamma^F$	9	167	99864
11	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^F \gamma^F \gamma^F$	12	63	99927
12	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^I \gamma^F \gamma^F$	10	61	99988
13	$e^+ e^- \rightarrow vpho \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^F \gamma^F \gamma^F$	11	4	99992
14	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^I \gamma^F \gamma^F \gamma^F$	15	4	99996
15	$vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^F \gamma^F \gamma^F \gamma^F$	14	2	99998
16	$vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^F \gamma^F \gamma^F \gamma^F$	13	1	99999
17	$e^+ e^- \rightarrow vpho \gamma^I \gamma^I, vpho \rightarrow \pi^- \bar{n} p \gamma^F \gamma^F \gamma^F \gamma^F$	$\pi^- \bar{n} p \gamma^I \gamma^I \gamma^F \gamma^F \gamma^F \gamma^F$	16	1	100000

Figure 4.2: Topology Analysis (TopoAna) output of the generated 100k events. The first column denotes the row number, the second the decay tree (decay chain), the third the decay final state, the fourth the unambiguous decay tree index, the fifth the number of entries per channel, and the last the cumulative number of entries

It is interesting to note that events without ISR emission are the most frequently generated ($\sim 35\%$), followed by those with two ($\gamma^I \gamma^I$) ($\sim 23\%$) and one (γ^I) ISR photon ($\sim 19\%$).

The $vpho$ (virtual photon) is a fictitious particle used by the generator Phokhara to produce ISR photons before the $e^+ e^-$ collisions and then enforce (evt_gen) the selected decay channel, approaching the following scheme:

$$\begin{aligned}
 e^+ + e^- &\rightarrow vpho + \gamma_{ISR} \text{ (powered by Phokhara generator)} \\
 vpho &\rightarrow p + \pi^- + \bar{n} \text{ (powered by evt_gen generator)}
 \end{aligned}$$

Additionally, numerous Final State Radiation (FSR) photons are emitted by the charged final state particles due to next order perturbative QED processes.

4.1.1 Applied selections and cuts

In the ISR case the reconstructed particle are $p\pi^-\bar{n}\gamma_{ISR}$ and the recoil system is made of $p\pi^-\gamma_{ISR}$, while in the NO ISR case the reconstructed particle are

$p\pi^-\bar{n}$ and the recoil system is made of $p\pi^-$.

In order to achieve a proper selection to select the signal in the recoil mass distribution, several cuts and Monte Carlo selections are applied within the steering file as follow:

1. Recoil system: $0 \text{ GeV}/c^2 < \text{recoil mass} < 2 \text{ GeV}/c^2$
2. $\bar{n}$: neutral clusters from ECL
3. p : $\text{protonID} > 0.9$ and $dr < 1 \text{ cm}$ and $|dz| < 3 \text{ cm}$
4. π^- : $\text{pionID} > 0.1$ and $dr < 1 \text{ cm}$ and $|dz| < 3 \text{ cm}$

The first one is a cut that selects the region of interest in the energy spectrum, corresponding to the interval $[0,2] \text{ GeV}/c^2$ around the antineutron mass ($\sim 939,56 \text{ MeV}/c^2$) with the positive mass constraint.

In order to analyze only $\bar{n}$ signals from the calorimeter, the second selection requires that the neutral cluster associated to a $\bar{n}$ candidate originates from the electromagnetic calorimeter (ECL).

The third and fourth criteria are standard selections recommended by the basf2 framework. The ID requirement imposes a constraint on the protonID or pionID based on a maximum likelihood criterion as follows:

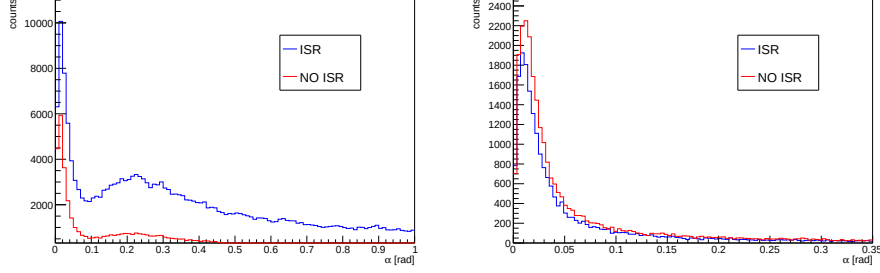
$$\text{protonID} = \frac{L_p}{L_e + L_\mu + L_\pi + L_K + L_d} \quad \text{pionID} = \frac{L_\pi}{L_e + L_\mu + L_p + L_K + L_d}$$

The cuts on dr and $|dz|$ select only tracks originating from the interaction point (IP).

Further cut is imposed using the basf2 **rankByLowest** method. It allows to select only the highest rank candidates, based on the angle between the neutral cluster and the recoil vector. In order to achieve a proper selection over $\bar{n}$ candidates, rankByLowest method is used (Appendix A). Since the recoil vector can have several neutral cluster, in $\bar{n}$ list candidates, associated with it, only the $\bar{n}$ candidates with the smallest angle α relative to the recoil is selected. Using spherical coordinate system, after some algebraic steps, the 3D angle α between the two vectors can be obtained:

$$\alpha = \arccos[\sin(\theta_1)\sin(\theta_2)\cos(\phi_1 - \phi_2) + \cos(\theta_1)\cos(\theta_2)]$$

Here vector **1** denotes the recoil vector and the vector **2** denotes the $\bar{n}$ direction vector. The formula is, of course, symmetric under the exchange $1 \leftrightarrow 2$. The respective distributions, before and after the aforementioned selection, are shown below in both ISR and NO ISR cases:

Figure 4.3: (a) α distribution (b) α distribution after the rank selection

In order to have a cleaner sample, only the candidates with $\alpha < 0.35$ ($\sim 20^\circ$) are stored. After the cut and the ranking selection, the α distribution is scattered in the interval $[0, 0.35]$ rad.

20146 candidates are reconstructed for the ISR case and 25013 for the NO ISR case, leading to an efficiency respectively of:

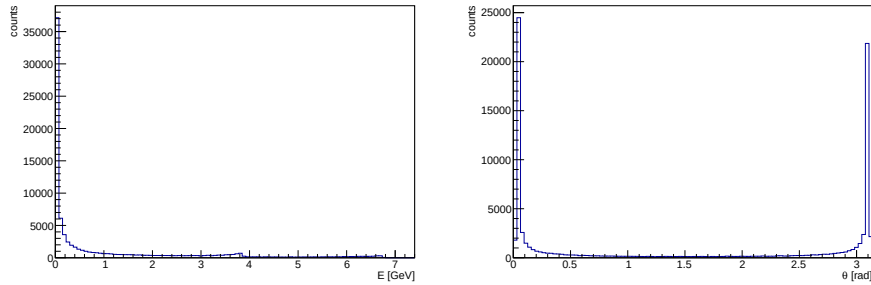
$$\epsilon = \frac{n^\circ \text{ of reconstructed candidates}}{n^\circ \text{ of generated events}} \sim 20\% \text{ and } 25\%$$

Since most of the ISR radiation is emitted in the backward ($\theta \sim 180^\circ$) and in the forward ($\theta \sim 0^\circ$) directions -where detection coverage is absent- the efficiency in the ISR case is slightly lower than in the NO ISR case. The ISR distributions are described below.

4.1.2 ISR distributions

Since ISR emission is independent of the analyzed channel, its energy and angular distributions are preliminary discussed. The lowering of the center-of-mass energy is the main effect of ISR emission, which leads to energy below the $\Upsilon(4S)$ resonance.

Therefore, the relative distributions are shown below:

Figure 4.4: (a) ISR energy distribution (b) ISR θ distribution

The ISR energy spectrum mainly consists of photons in the $[0, 1]$ GeV range. The angular distributions present a U-shape, indicating that forward and backward directions are the most probable, as anticipated before. Since electrons (7 GeV beam) go towards forward direction and positrons (4 GeV beam) go towards backward direction, ISR θ distribution reflects the intrinsic forward Lorentz boost which helps the $B^+ B^-$ (or $B_0 \bar{B}_0$) secondary vertexes identification.

Furthermore, the peak at low θ values (forward direction) is higher than the backward peak, consistent with the detector asymmetry with respect to the interaction point.

4.1.3 The recoil mass

It is necessary to check that the recoil system $p\pi^-(\gamma_{ISR})$ is properly described. Specifically, the reconstructed recoil mass should peak at the nominal anti-neutron mass, since no background channels are present in this sample. The only background consist of secondary particles produced during the propagation through the apparatus. The result for both ISR and NO ISR cases is shown:

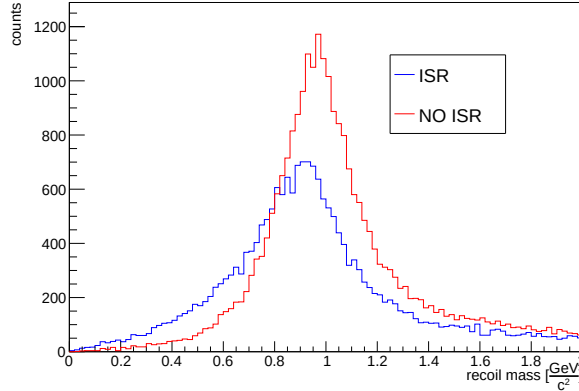


Figure 4.5: Recoil mass distribution, in both ISR and NO ISR cases

It is easy to notice that the distribution peaks above the $\bar{n}$ mass (~ 939.56 MeV), indicating that the the recoil mass is well reconstructed in both ISR ($p\pi^-\gamma_{ISR}$) and NO ISR ($p\pi^-$) cases. Therefore the variables associated with the $\bar{n}$ clusters can be compared with the reconstructed recoil variables (p, θ). As said before, there is more than one γ_{ISR} per event. Then the [blue distribution](#) shows more entries in the left tail, since only the most energetic gamma is involved in the recoil reconstruction, while the others are neglected. On the other hand, since many ISR photons end up in the forward or backward regions with poor detector coverage, the [red distribution](#) exhibits a higher peak as the recoil system reconstruction is only made of $p\pi^-$.

4.1.4 The recoil kinematic

Before analyzing the recoil momentum distribution, it may be useful to examine the reconstructed and generator-level momenta for each particle which compose the recoil system. The relative reconstructed momenta are shown below:

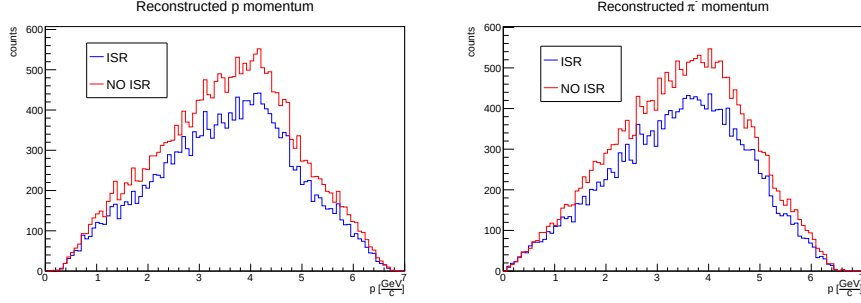


Figure 4.6: Reconstructed momentum distribution of proton and pion, in both ISR and NO ISR cases

The distributions clearly exhibit the typical phase-space shapes, as expected from the adopted generator model (Appendix A). Similarly, these distributions can be obtained directly at the generator level, yielding:

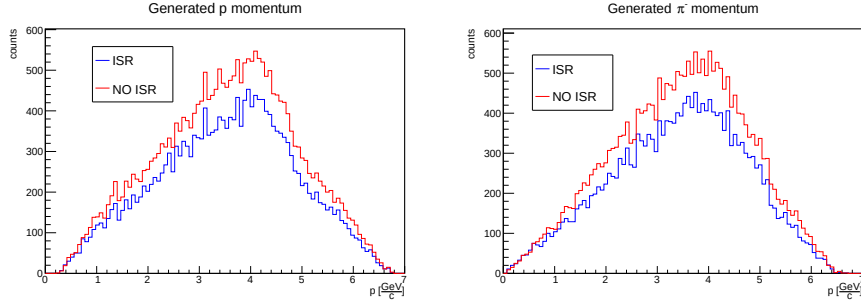


Figure 4.7: Generated momentum distribution of proton and pion, in both ISR and NO ISR cases

Both distributions exhibit similar shapes, indicating a proper reconstruction of charged particles by the software. The reconstructed recoil momentum can be obtained in both ISR and NO ISR cases, as follows:

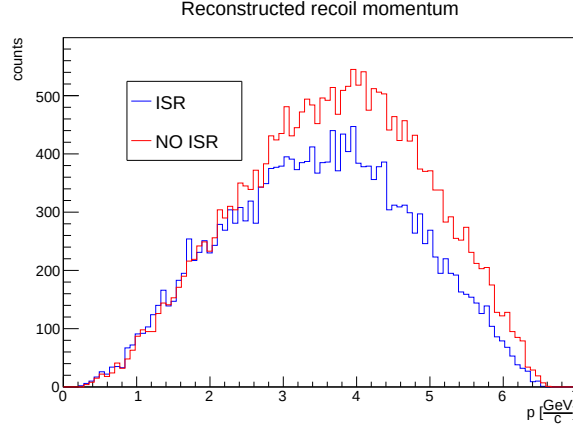


Figure 4.8: Reconstructed momentum distribution of the recoil system, in both ISR and NO ISR cases

Since the recoil system is well reconstructed through the recoil mass, it could be interesting to compare this distribution with that of $\bar{n}$, both from reconstruction and from generator. The latter is not affected by detector propagation and therefore it represents the $\bar{n}$ candidates generated momentum, free from misidentification and directly originating from the generator. The aforementioned $\bar{n}$ distributions are shown below:

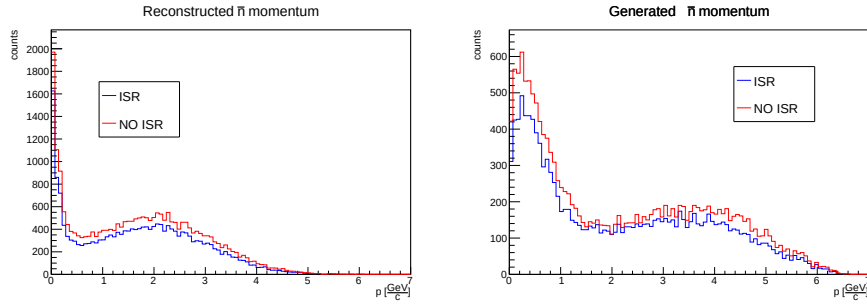


Figure 4.9: Reconstructed (left) and generated (right) momentum distribution of anti-neutron candidates, in both ISR and NO ISR cases

Notably, the reconstructed $\bar{n}$ momentum is systematically underestimated compared to the generator-level one, indicating missing energy within neutral clusters. This topic will be addressed in detail below through the momentum residuals.

Both the reconstructed and generator-level distributions exhibit two structures, indicating two contributions within $\bar{n}$ candidate neutral clusters. To investigate this, the reconstructed recoil momentum is compared with the generator-level momentum of all $\bar{n}$ candidates, using MC truth matching. Since the $\bar{n}$ PDG

code is **-2112**, the neutral clusters associated to $\bar{n}$ list can be matched to **true** (truth = 1) $\bar{n}$ particles, using Monte Carlo PDG = -2112, or to **false** (truth = 0) candidates with Monte Carlo PDG $\neq$ -2112:

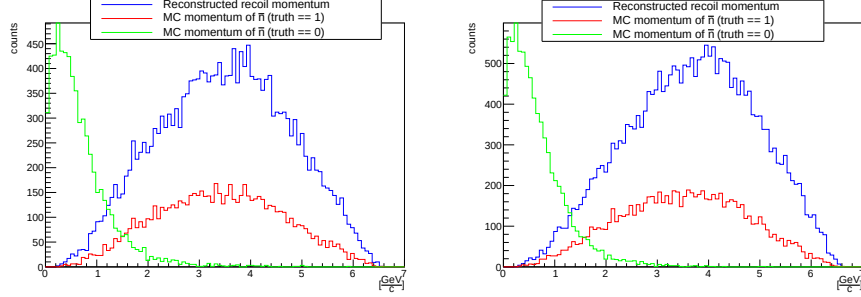


Figure 4.10: (left) Momentum contribution component in ISR case
(right) Momentum contribution component in NO ISR case

As expected, the low-momentum structure corresponds to candidates not associated to $\bar{n}$ by the Monte Carlo truth, exhibiting significant deviation from the recoil momentum. This is likely due to cluster split-off and/or pre-showering effects as will be discussed later. Furthermore, the **red distribution** is include within the **blue one**, suggesting that several anti-neutron are missing after the reconstruction. All these aspects will be discussed in the next section through analysis of $\bar{n}$ relatives (ancestors).

To compare the reconstructed recoil with the Monte Carlo matched $\bar{n}$ (truth = 1), it is useful to analyze their linear correlation coefficient ρ , recalling that:

$$\rho_{x,y} = \frac{cov[x,y]}{\sigma_x \sigma_y} \quad \text{where } cov[x,y] = E[xy] - E[x]E[y]$$

Therefore, the correlations in θ , for both the reconstructed and the generated case, are shown below:

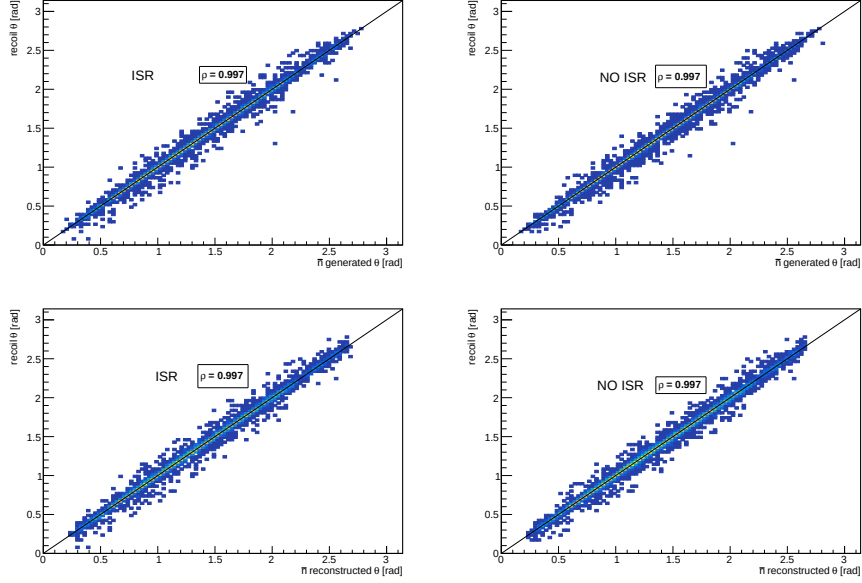


Figure 4.11: θ correlation distribution between recoil system and anti-neutron candidate, in both reconstructed and generated case. Linear correlation coefficient ρ truncated to three decimal places

A good correlation can be highlighted between the two distributions in both reconstructed and generator-level cases, indicating that the the anti-neutron θ direction is properly described.

The same comparison can be addressed for the momentum, remembering that it is obtained starting from the energy of the cluster as follow:

$$p_{rec}^2 = E_{ECL}^2 - m_n^2$$

The obtained conjugate distributions are shown below:

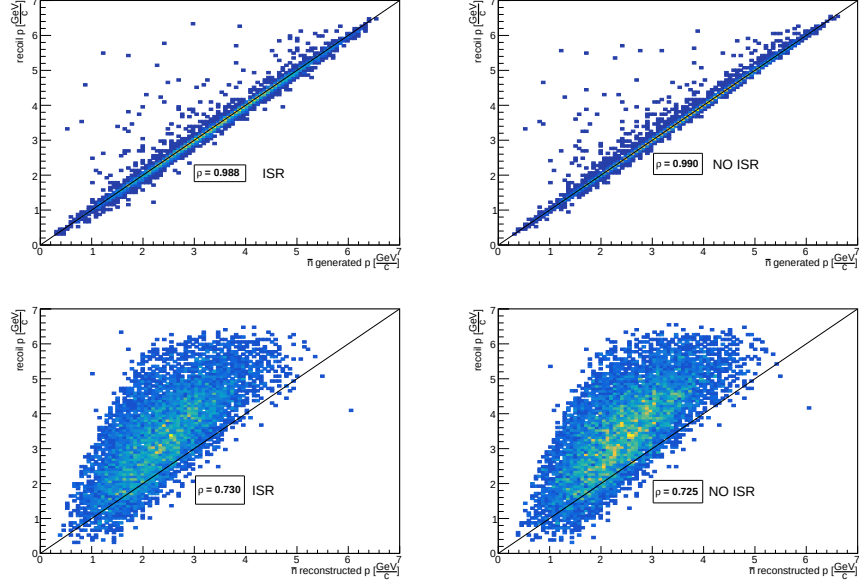


Figure 4.12: Momentum correlation distribution between recoil system and anti-neutron candidate, in both reconstructed and generated case. Linear correlation coefficient ρ truncated to three decimal places

A good correlation can be observed between the reconstructed recoil momentum and the generated $\bar{n}$ one, confirming how the first is properly describing the $\bar{n}$ kinematic.

However, a poor resolution is observed in reconstructed $\bar{n}$ momentum, in both ISR and NO ISR cases, presenting an underestimation of the reconstructed $\bar{n}$ momentum as well. In order to directly study it, the momentum residual can be analyzed:

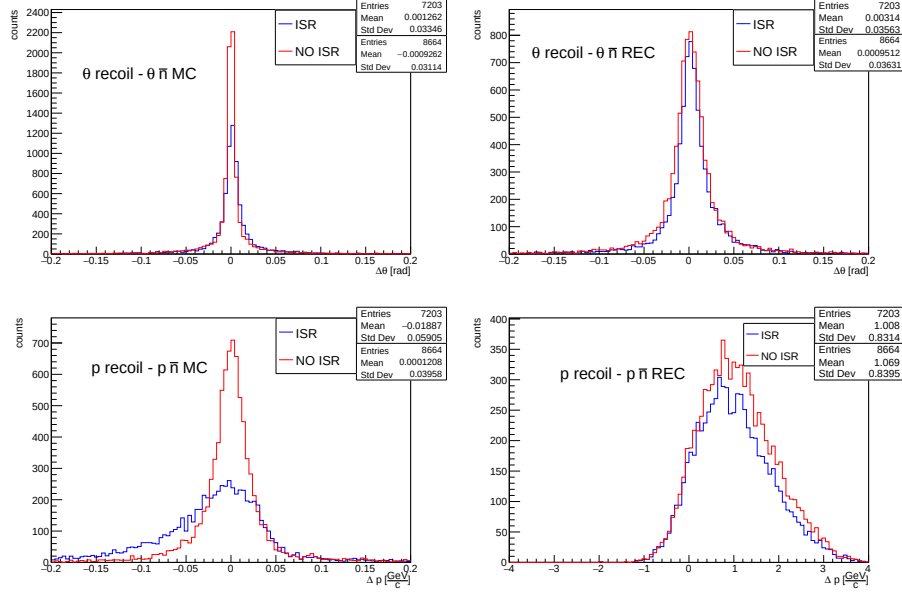


Figure 4.13: Momentum and θ angle residuals distribution between recoil system and anti-neutron candidate

Besides exhibiting poor resolution, the reconstructed momentum of the $\bar{n}$ is underestimated as the residuals distributions show, suggesting a missing energy in the shower. Therefore, the $\bar{n}$ momentum is poorly reconstructed using only the ECL information. Most of anti-neutrons undergo in annihilation with hadrons such as $\bar{n}n$ and $\bar{n}p$, producing pions which escape the calorimeter. Therefore, the reconstructed momentum (or energy) in the calorimeter is not directly comparable with the one of anti-neutrons.

4.2 The $\bar{n}$ ancestors analysis

As observed, cluster split-off and/or pre-showering effects can occur distorting the $\bar{n}$ reconstructed distributions.

Since the ISR and NO ISR results are analogous, only the ISR case are analyzed on the following section for smoother reading.

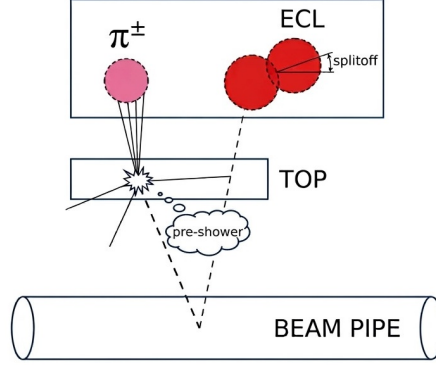


Figure 4.14: Artistic sketch displaying cluster split-off and pre-showering effects following $\bar{n}$ annihilation. Cluster split-off occurs in the ECL, while pre-showering takes place in the TOP; the respective products then travel toward the ECL and then mis-identified as the primary cluster associated to the anti-neutron

The secondary particle from $\bar{n}$ annihilation can be detected as the primary ECL cluster, resulting in information loss and requiring the TOP time information to improve the anti-neutron detection.

4.2.1 The ancestor meta-variables

In order to study the chain and the kinship degree of the neutral cluster associated to $\bar{n}$ candidates detected in the ECL, the basf2 **ancestor** meta-variables are analyzed, such as:

- **varForFirstMCAncestorOfType(type, variable)**: this metavariable returns the requested **variable** of the first ancestor of the given **type** -a specific particle- found in a certain point of the chain
- **hasAncestor(PDG, abs)**: this metavariable returns a positive integer if an ancestor -a certain relatives in the chain- with the given **PDG** code is found, 0 otherwise. It means that 1 = first mother, 2 = grandmother, etc. Second argument **abs** is optional: 1 means that the sign of the PDG code is taken into account, default is 0

A typical scenarios of $\bar{n}$ as ancestor could be the following:

- 0) $\bar{n}$ from IP $\rightarrow$ 1) $\bar{n}$ annihilation in TOP (pions produced) $\rightarrow$ 2) pions detected
- 0) $\bar{n}$ from IP $\rightarrow$ 1) $\bar{n}$ annihilation in ECL (pions produced) $\rightarrow$ 2) pions detected

Therefore the variables **varForFirstMCAncestorOfType(anti-n0, p)** and **hasAncestor(-2112, 1)** of neutral cluster associated to $\bar{n}$ are discussed. Given the close similarity between the results, only the ISR case is discussed. The hasAncestor distribution is shown below:

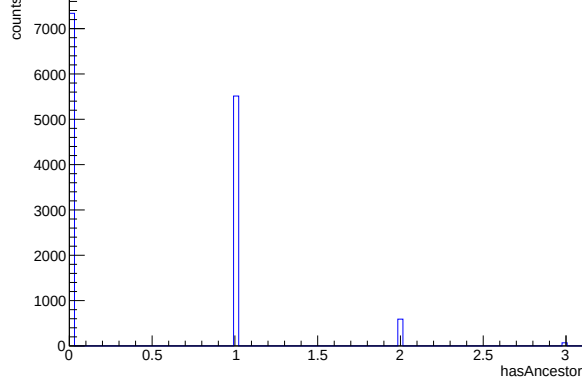


Figure 4.15: Distribution of the ancestry kinship degree of anti-neutron candidates with an anti-neutron present in their ancestry chain

Consequently, most $\bar{n}$ candidates lack an $\bar{n}$ in their relative chain ($\text{has_Ancestor}=0$), suggesting they are the primary $\bar{n}$ themselves. The second contribution is made of $\bar{n}$ candidates having $\bar{n}$ as mother etc.

In order to directly study the contribution of each term, it could be useful to access to the Monte Carlo truth and look at the PID trough the **mc.PDG**, which basically is the real PDG of the selected candidates coming from the generator. The obtained distributions are shown below:

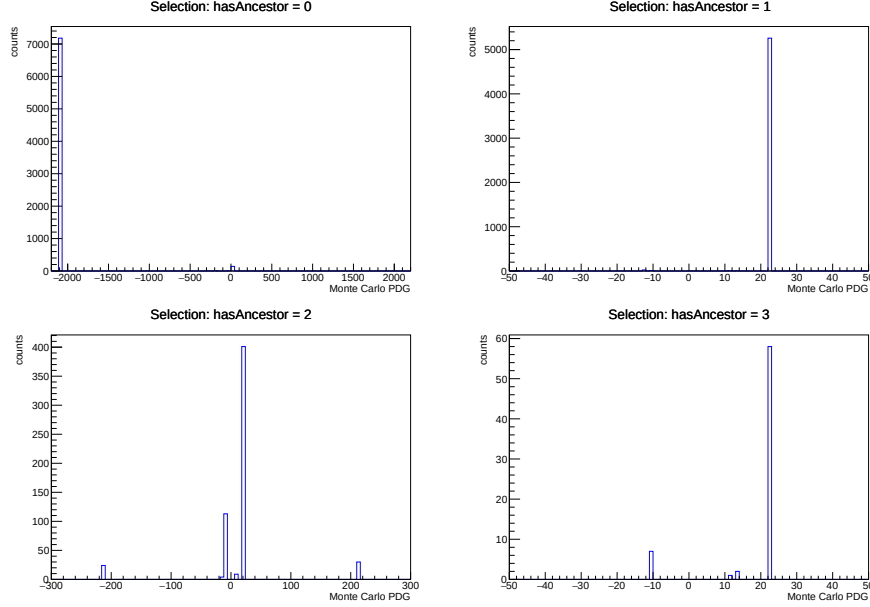


Figure 4.16: Monte Carlo PDG codes of candidates with degrees of ancestry selection. As mentioned, the MC PDG variable reveals the generator-level identity of the reconstructed particle by providing its PDG code

- **hasAncestor = 0** clearly shows that $\bar{n}$ candidates without an $\bar{n}$ as relatives are mainly true anti-neutrons, as their Monte Carlo PDG code is primarily -2112. This is consistent since the anti-neutron comes from the IP and it is directly detected in the ECL through its own annihilation
- **hasAncestor = 1** shows that the $\bar{n}$ candidates with an $\bar{n}$ as mother are predominately gammas (Monte Carlo PDG = 22), suggesting they result from $\bar{n}$ interactions prior to $\bar{n}$ candidate ECL cluster formation
- **hasAncestor = 2** suggests that the $\bar{n}$ candidates with an $\bar{n}$ as grandmother are mainly gammas, with some spurious pions (-211 & +211) and electrons or positrons (+11 & -11). The case with hasAncestor = 3 yields analogous results

4.2.2 Ancestor momentum

The figure 4.10 can be updated, by including the momentum of the ancestors obtained with `varForFirstMCAncestorOfType(anti-n0, p)`. The idea is to check if the information from the first structure -the one at low value of momentum- can be recovered using the ancestors chain. The obtained distributions are shown below:

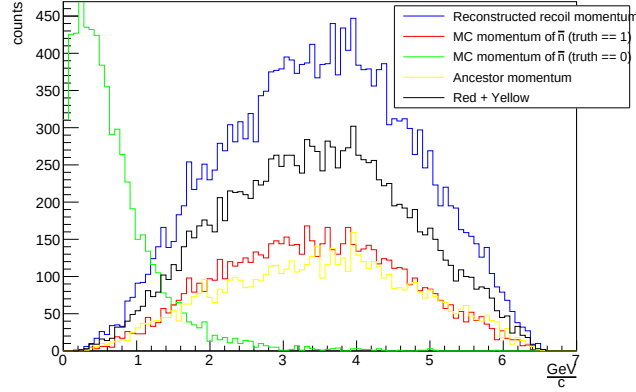


Figure 4.17: Momentum comparison of different components. The recoil vector is the only reconstructed variable, while the others are obtained from Monte Carlo truth selection, as shown in the legend

The **blue distribution** is made of 20146 entries, the **red one** is made of 7203 entries, the **green one** is made of 6308 entries, the **yellow one** is made of 6173 entries and the black one is made of 13376 entries. It means the 98% of the first structure is recovered by ancestor momentum, suggesting that phenomena prior to neutral cluster associated to $\bar{n}$ candidates can occur, such as cluster split-off and pre-showering. The quality of the ancestor momentum can be discussed by studying the residuals with respect to the reconstructed recoil momentum:

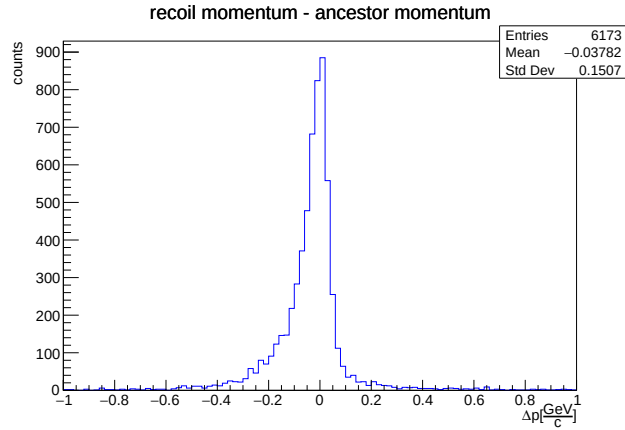
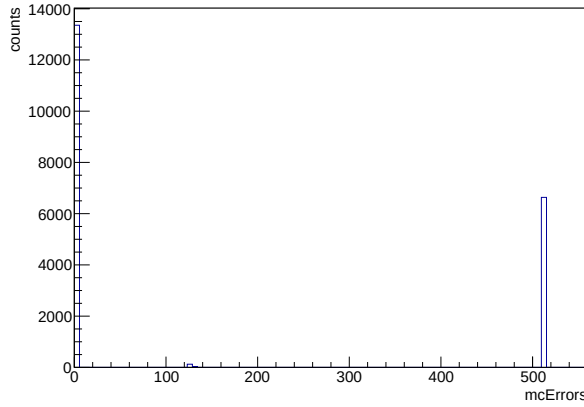


Figure 4.18: Residual distribution of ancestor momentum with respect to the recoil vector

Therefore the reconstructed recoil momentum is a good estimator of the

ancestor momentum as well, confirming that cluster split-off or pre-showering may occur prior the ECL cluster formation.

It is interesting as well to note that $20146 - 13376 = 6779$ which exactly matches the number of $\bar{n}$ candidates associated with a **NaN** value, mainly particles that did not receive a correct Monte Carlo association during MC truth matching. They cannot be recovered. The meta-variable **mcErrors** shows as a bit number the reason why the Monte Carlo fails the association. This variable explicitly shows this behavior:

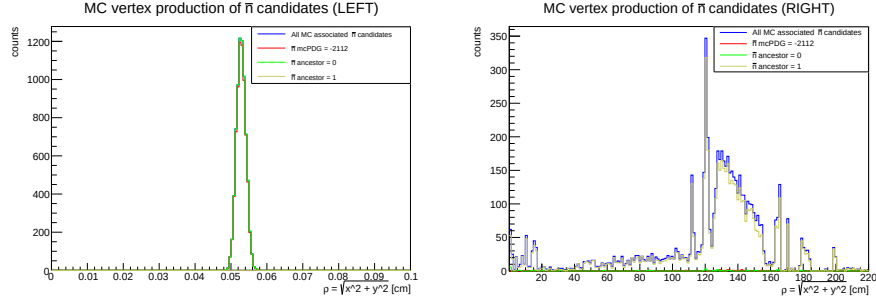


Where the first peak is above the code **0**, which means that " *This particle and all its daughters are perfectly reconstructed*", while the second peak is above the code **512**, which confirms that " *There was an error in MC matching. Not a valid match. Might indicate fake/background track or cluster*", as expected.

4.2.3 $\bar{n}$ vertex production

Eventually, in order to a comprehensive ancestor analysis, it could be useful to discuss the generator-level radial production distance $\rho = \sqrt{x^2 + y^2}$ -the vertex production- of $\bar{n}$ candidates, from the z-axis.

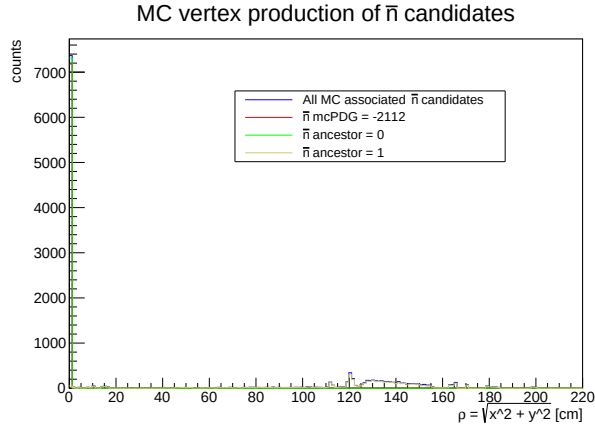
To provide a clear overview of ρ , the distribution is splitted in two parts: the left part -close to the collision diamond- ($\rho \in [0, 0.1]$ cm) and the right part -far from the collision diamond- ($\rho \in (0.1, 220]$ cm). The obtained result is shown below:


 Figure 4.19: $\bar{n}$ production vertex: (left) close to the IP (right) far from the IP

As expected, the correctly associated anti-neutrons (**red distribution**) overlay the $\bar{n}$ candidates without an anti-neutron ancestor (**green distribution**) and they are from the interaction point (left part).

As seen before, the $\bar{n}$ candidates with a $\bar{n}$ as mother are mainly photons and they are produced around the detector (**yellow distribution**). It is interesting to note that the first structure is in the vertex detectors region ($\rho < 16$ cm) and the two peaks located at the TOP edges, mainly around 118 cm (where the TOP begins) and 125 cm (where the TOP ends and the ECL starts), explicitly showing how cluster split-off and pre-showering events occur in the events.

To compare the height comparison between the left and right contributes, the full distribution is shown below:


 Figure 4.20: $\bar{n}$ production vertex showing the IP peak is the main one

4.3 Kinematic Fit effects

A **Kinematic Fit** is a constrained optimization technique used in particle physics to refine the measured four-momenta (or other kinematic quantities)

of event particles by imposing known physical constraints, such as energy-momentum conservation or recoil mass. Measured quantities like track momenta, energies from calorimeters, or decay vertices come with resolution-limited errors from detectors. The fit adjusts these within their error bars to best satisfy the imposed constraint.

There are several type of Kinematic Fit that could yield improved resolution. The one constraint **1C Kinematic Fit** has been tested over the $p\pi^-(\gamma_{ISR})$ system recoil mass, forcing it to the anti-neutron mass (~ 0.9385 GeV) and then studying the response on the recoil kinematic variables resolution versus the anti-neutron candidate ones. The obtained residuals are shown below:

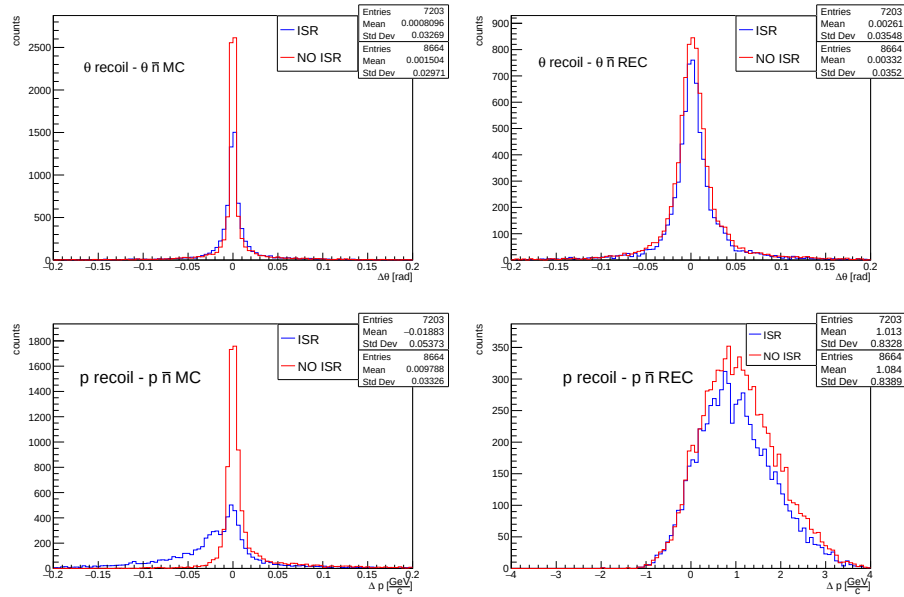


Figure 4.21: Momentum and θ angle residuals distribution between recoil system and anti-neutron candidate, after the 1C Kinematic Fit procedure

As can be easily seen from the distributions, no significant improvement in recoil momentum resolution was observed post 1C Kinematic Fit compared to the unconstrained baseline 4.12. This confirms that the recoil system was already optimally reconstructed in the signal sample Monte Carlo. Further kinematic fit studies using a Monte Carlo cocktail are discussed in the next section.

4.4 Shower shape variables in the signal sample

The following chapter introduces the core of this thesis: the analysis of the **shower shape variables** of neutral ECL clusters associated with anti-neutron

candidates. Since the final goal is to assess Data/MC agreement for these variables, the shower shape variables are first studied in an MC signal sample with no background channels present. The main idea is that particle identification can be achieved by studying the shower shapes and their characteristics.

4.4.1 The analyzed shower shape variables

The selected variables are the **cluster energy**, the **E1E9** variable, the **E9E21** variable, **Lateral Momentum**, **Second Moment**, **Zernike Moments** (50 and 41), and **Number of Hits**, which are described first.

Cluster Energy

The Cluster Energy (*clusterE* in the basf2 software), as the name suggests, represents the total energy deposited by the cluster associated with a given candidate particle, in this case the anti-neutron. It has units of GeV and only clusters with Cluster Energy above 20 MeV are stored by the reconstruction software.

Cluster E1E9 and E9E21 In order to provide a proper definition of these variables (respectively *clusterE1E9* and *clusterE9E21* in the basf2 software), the following figure is shown below:

10	11	12	13	14
25	2	3	4	15
24	9	1	5	16
23	8	7	6	17
22	21	20	19	18

Figure 4.22: Ideal representation of a cluster pattern, consisting of a 5×5 matrix in which each cell corresponds to a single crystal

Cluster E1E9 and cluster E9E21 are defined as follow:

$$E1E9 = \frac{E_1}{\sum_{i=1}^9 E_i} \quad E9E21 = \frac{\sum_{j=1}^9 E_j}{\sum_{j=10}^{25} E_j}, \text{ with } j \neq 10, 14, 18, 22$$

The first one returns the ratio of the energy in the central crystal (E_1) to the total energy in the surrounding 3×3 crystal grid (E_9). The second one returns the ratio of the energy in the 3×3 crystal grid around the central crystal (E_1)

to the total energy in the 5×5 crystal grid around the central crystal, excluding the corners.

These ratios are always less than 1 and tend toward larger values for photons and smaller values for hadrons, as the latter produce more spread-out showers. These variables are dimensionless and are included within the range $[0, 1]$.

Lateral Momentum

The cluster Lateral Momentum or Lateral Energy distribution (*clusterLAT* in the basf2 software) is defined as follow:

$$C_{LM} = \frac{\sum_{i=2}^n E_i r_i^2}{E_0 r_0^2 + E_1 r_1^2 + \sum_{i=2}^n E_i r_i^2},$$

with $r_0 \simeq 5$ cm (average distance between two crystals centres).

n is the number of crystals associated to the shower, $E_i = (E_0, E_1, \dots)$ are the crystal energies sorted by descending energy and r_i is the distance of the i -th digit to the shower center projected onto a plane perpendicular to the shower axis. For radially-symmetric electromagnetic showers, this variable peaks at ~ 0.3 , where it is larger for hadronic particles. This variable is dimensionless and it is included within the range $[0, 1]$.

Second Moment

The cluster Second Moment (*clusterSecondMoment* in the basf2 software) is defined as follow:

$$C_{SM} = \frac{1}{S_0} \frac{\sum_{i=0}^n E_i r_i^2}{\sum_{i=0}^n E_i},$$

where E_i and r_i are the same variables as used for the lateral momentum, while S_0 is a normalization factor that normalizes to 1 for isolated photons, making C_{SM} dimensionless. The Second Moment measures the dispersion of cluster size.

Zernike Moments

The Zernike Moments (Zernike Polynomials) are a set of orthogonal functions which are defined on the unit disk. Since each moment represents a specific pattern shape, they are widely used to represent smooth, two-dimensional functions (especially wavefronts) over circular apertures.

On the unit disk parametrized by polar coordinates (ρ, ϕ) with $0 \leq \rho \leq 1$, the Zernike modes are usually written as:

$$Z_n^m(\rho, \phi) = R_n^m(\rho) \cos(m \phi)$$

$$Z_n^{-m}(\rho, \phi) = R_n^m(\rho) \sin(m \phi)$$

Where:

$$R_n^m(\rho) = \sum_{k=0}^{\frac{n-m}{2}} \frac{(-1)^k (n-k)!}{k! \left(\frac{n+m}{2}-k\right)! \left(\frac{n-m}{2}-k\right)!} \rho^{n-2k}$$

n represents the radial degree and m represents the azimuthal frequency ($m = 0$ for spherical Zernike polynomials). A figure of the Zernike polynomials is given below:

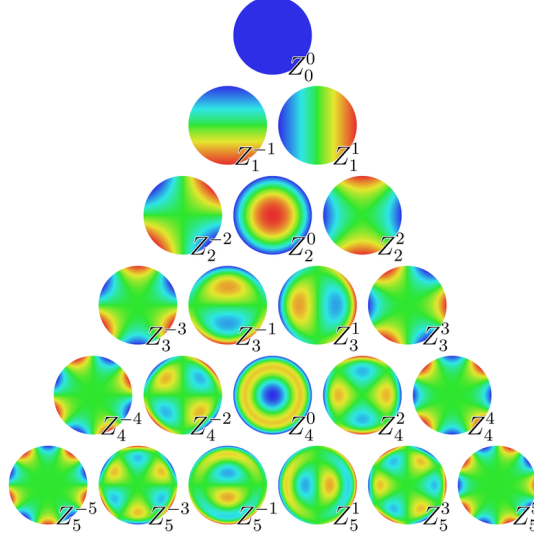


Figure 4.23: Zernike Polynomials representation. Moments Z_4^0 and Z_5^1 are widely discussed in this thesis

Precisely, Zernike Moments Z_4^0 and Z_5^1 (*clusterAbsZernikeMomen40* and *clusterAbsZernikeMoment51* in the basf2 software) are studied in this thesis, where the first one represents the radial spread distribution structure, while the second one represents the level of dipolar asymmetry.

Number of Crystals hit

The variable Number of Crystals hit (*clusterNhits* in the basf2 software) returns sum of weights w_i , with $w_i \leq 1$ of all crystals in the cluster. For non-overlapping clusters this is equal to the number of crystals in the cluster. However, in the case of overlapping clusters where individual crystal energies are split among them, this can be a non-integer value.

4.4.2 Anti-neutrons ECL cluster variables

In order to isolate the contribution of the neutral clusters which are correctly associated with $\bar{n}$ particles (Monte Carlo PDG = -2112) (**red distribution**) from all $\bar{n}$ candidates (**blue distribution**), the two distributions are overlaid for each cluster variable. The former is naturally included within the latter, naturally.

The obtained cluster variables in ISR case are shown below:

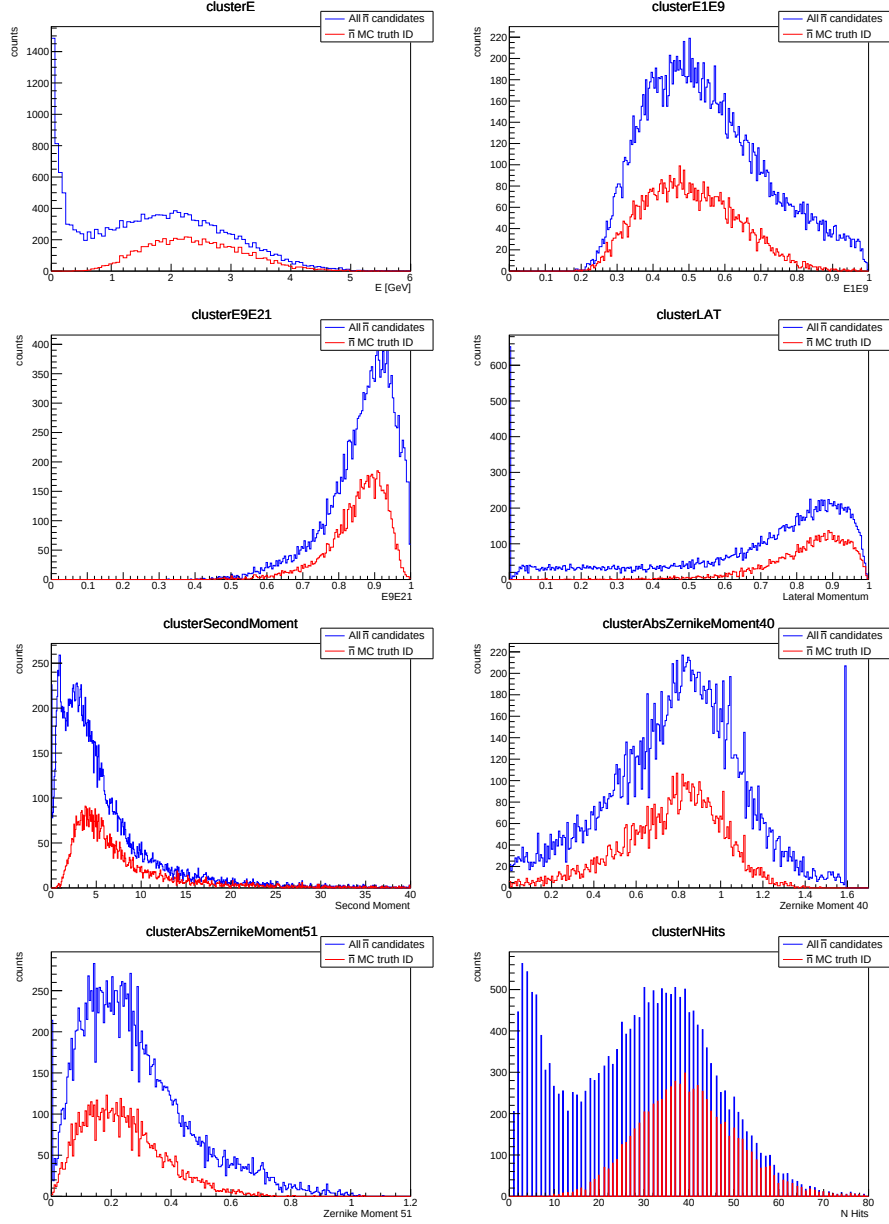


Figure 4.24: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates versus those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (ISR case). The selection is performed using the Monte Carlo PDG code (-2112), as previously discussed

The obtained cluster variables in NO ISR case are shown below:

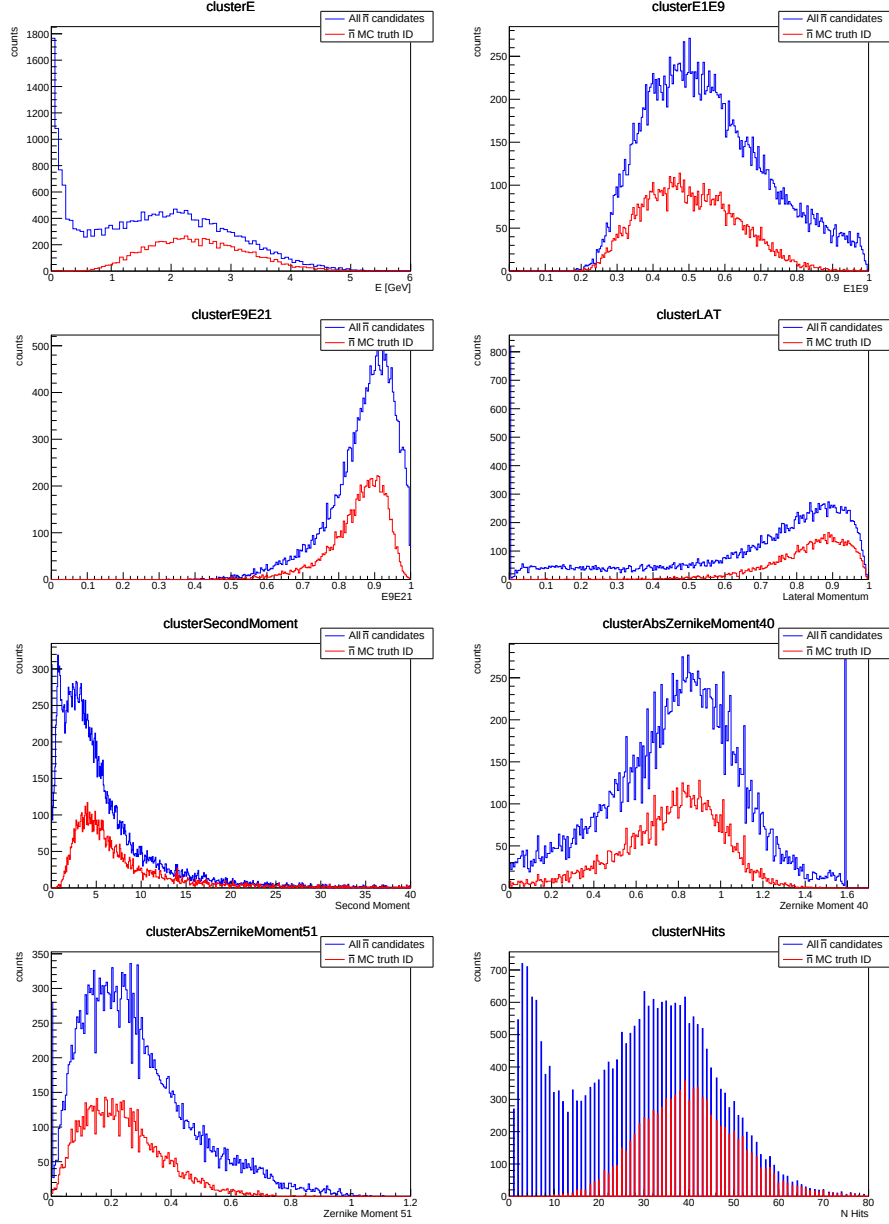


Figure 4.25: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates versus those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (NO ISR case). The selection is performed using the Monte Carlo PDG code (-2112), as previously discussed

First, it is important to highlight that the ISR and non-ISR distributions carry the same information, as expected. Indeed, the measurement of cluster variables is independent of the recoil system, which could be made of either $p\pi^-\gamma_{ISR}$ or $p\pi^-$ respectively.

It is also noteworthy that the [blue distribution](#) shows some peculiar structures respect the [red distribution](#). The reconstructed cluster energy for all $\bar{n}$ candidates exhibits a first structure at low energy values that resembles those discussed for the anti-neutron momentum distributions. This is also evident in the second moment distribution, where a low-energy structure is clearly visible. The most prominent difference appears in the number of hits ($N Hit$) distribution, where a distinct structure at $N Hit < 10$ is evident.

The nature of these characteristic structures is analyzed in the following subparagraph.

4.4.3 Ancestor cluster variables

It would be interesting to investigate the nature of this structure using the same approach as before, namely studying the ancestor contributions. The shower shapes with the following selections are compared:

- $\bar{n}$ Monte Carlo PDG = -2112 (neutral clusters correctly associated with $\bar{n}$ particles)
- $\bar{n}$ Monte Carlo PDG = NaN (neutral clusters not associated with any particles)
- $\bar{n}$ hasAncestor = 0 (neutral clusters with no $\bar{n}$ as relative)
- $\bar{n}$ hasAncestor != 0 (neutral clusters with $\bar{n}$ as relative)

Referring to figures 4.26 and 4.27, as expected the neutral clusters correctly associated with $\bar{n}$ particles carry the same information of those without $\bar{n}$ as relative, in both ISR and NO ISR cases.

Furthermore, it is evident how reconstructed neutral clusters not associated with any particles by the Monte Carlo and those with $\bar{n}$ as relative are primarily responsible for the structures observed at low values of energy, second moment, and number of crystals.

A further selection on number of crystals ($clusterNHits > 10$) can be performed to study the corresponding cluster variables. The result is shown in figure 4.28 and 4.29.

As observed, these structures arise from small-sized clusters produced by $\bar{n}$ particles that undergo phenomena -such as cluster split-off and pre-showering- prior to neutral cluster formation for $\bar{n}$ candidates, as confirmed by ancestor studies. The requirement of at least 10 crystals eliminates these structures.

The obtained cluster variables in ISR case are shown below:

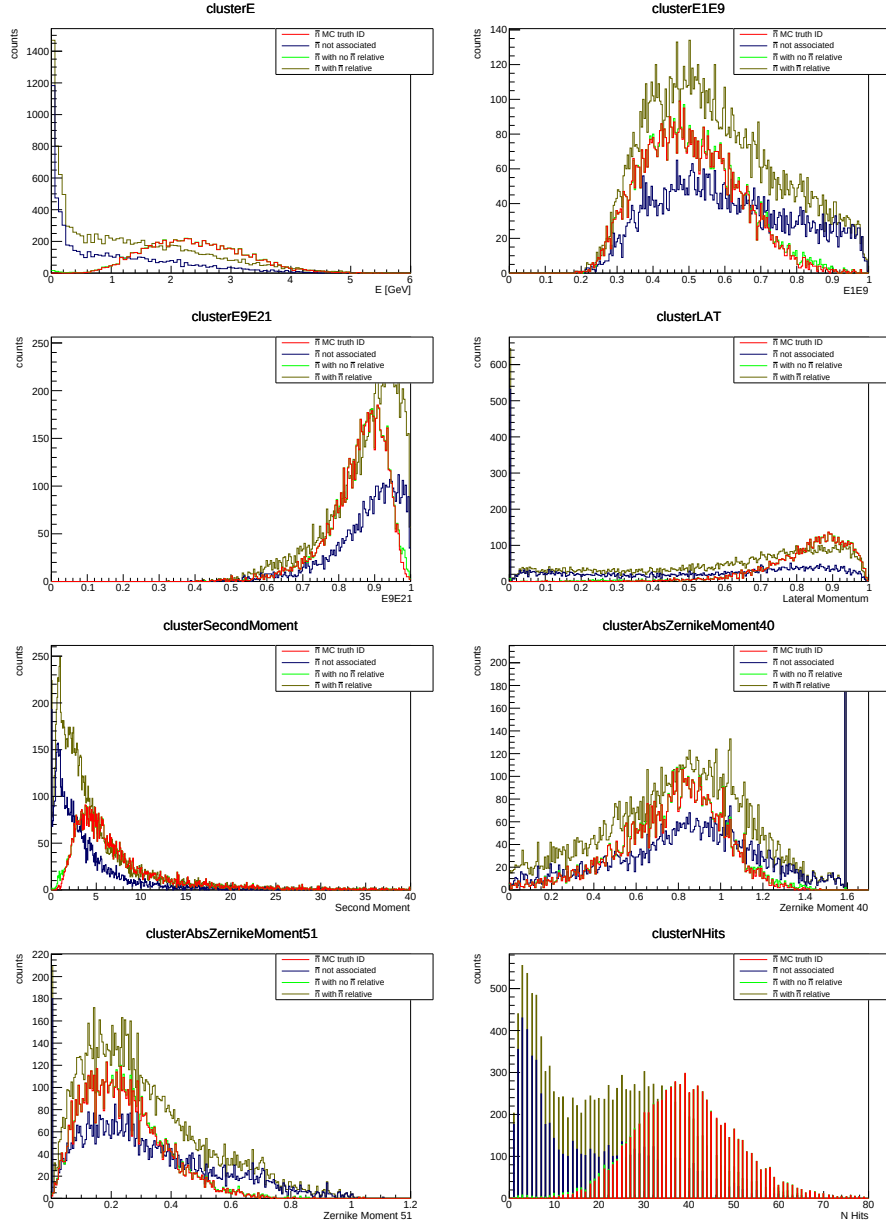


Figure 4.26: Comparison of reconstructed cluster variables for $\bar{n}$ candidates with different ancestor selection (ISR case)

The obtained cluster variables in NO ISR case are shown below:

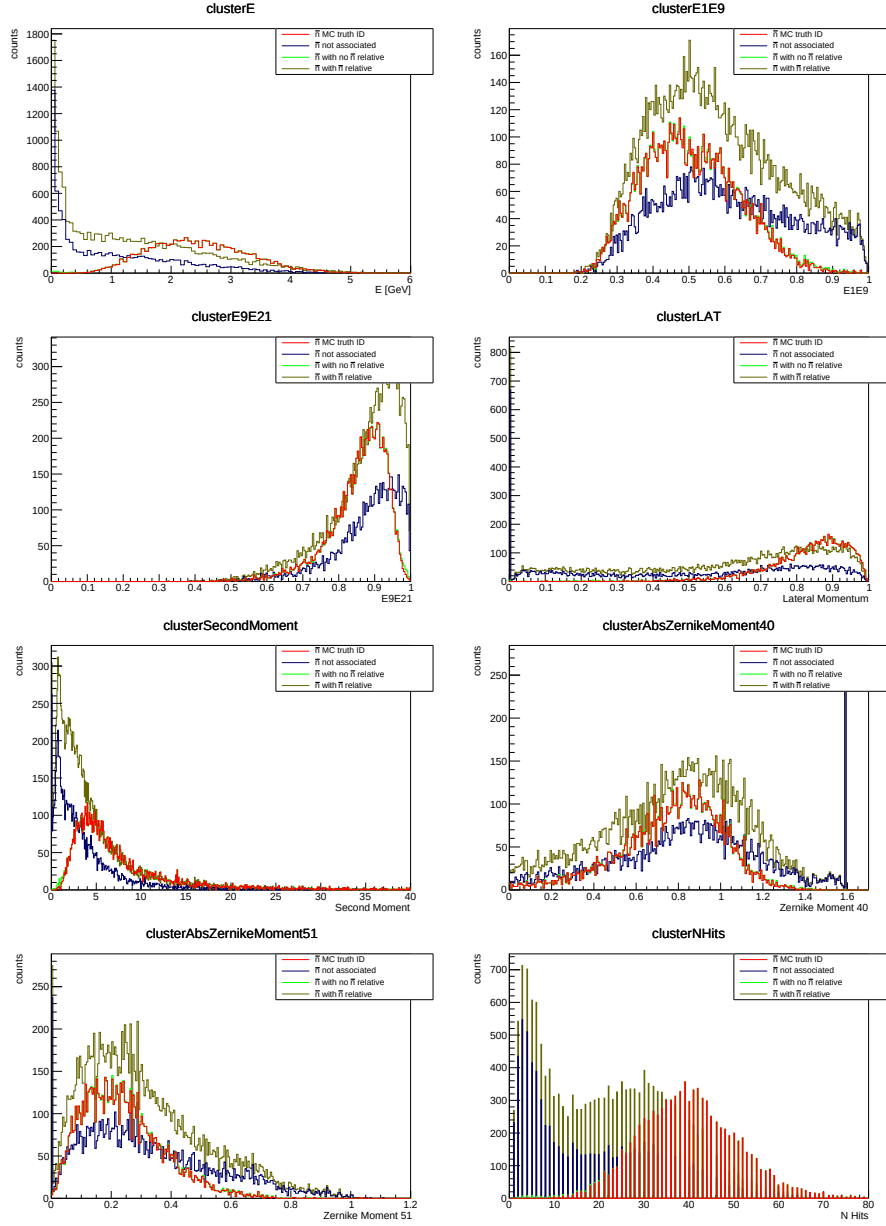


Figure 4.27: Comparison of reconstructed cluster variables for $\bar{n}$ candidates with different ancestor selection (NO ISR case)

The obtained cluster variables in ISR case are shown below:

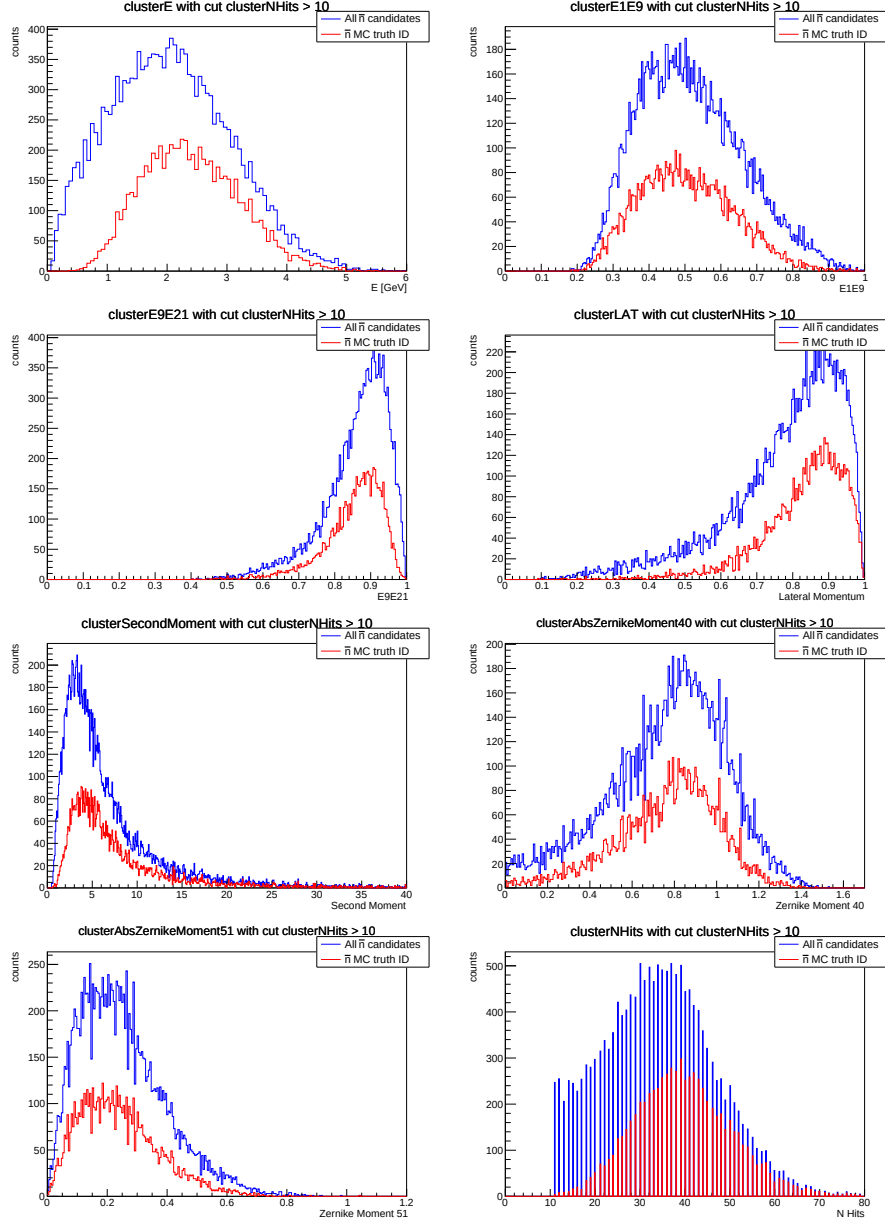


Figure 4.28: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates versus those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (ISR case). Cut $clusterNHits > 10$ is performed

The obtained cluster variables in NO ISR case are shown below:

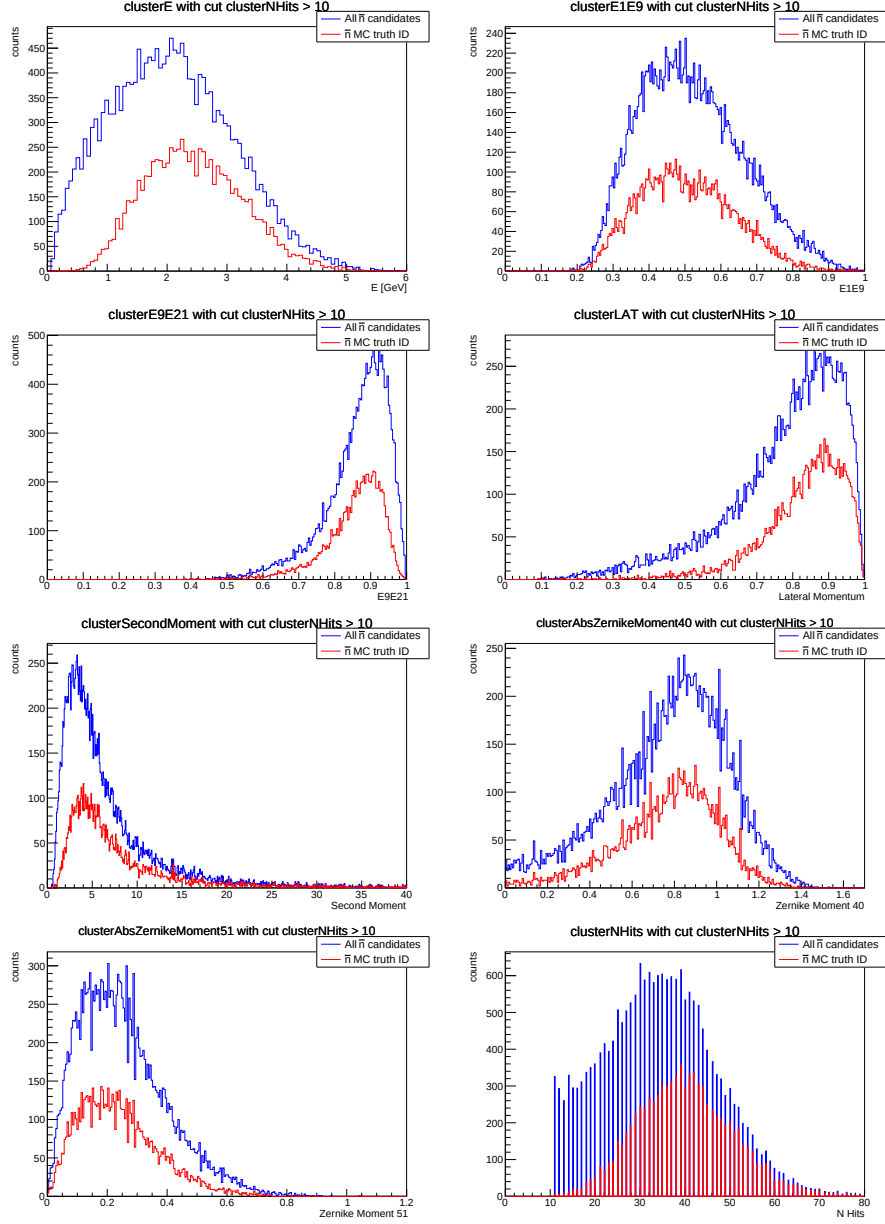


Figure 4.29: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates versus those associated with $\bar{n}$ particles by Monte Carlo truth Monte Carlo PDG = -2112 (NO ISR case). Cut $clusterNHits > 10$ is performed

4.4.4 Cluster variables in different ECL regions

As described in Chapter 3, the Belle II detector is divided into barrel and forward/backward end-cap regions. It would be interesting to study the reconstructed cluster variables in each ECL region separately. Since the ISR and NO ISR cases exhibit analogous shower shapes, only the latter is discussed here. The region hit distribution is shown below:

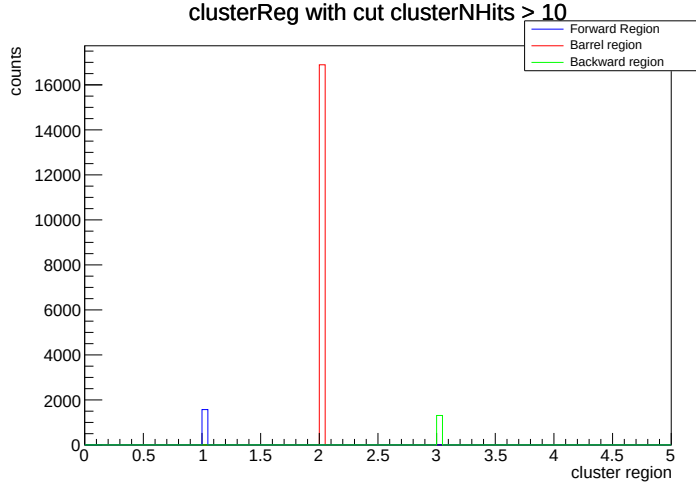


Figure 4.30: Different ECL regions contribution in neutral cluster associated to $\bar{n}$ candidates, where 1 is the forward region, 2 is the barrel region and 3 is the backward region. As expected, the barrel region is the most involved

As observed in Fig. 4.31, the cluster variable associated to all the reconstructed $\bar{n}$ shapes are consistent across all ECL regions.

However a slight difference can be noted for cluster energy and number of hits variables. In fact, these two exhibit a shift toward lower values in the forward region, reflecting the Lorentz boost of the center-of-mass. In any case, the barrel region contains the majority of ECL clusters, as expected.

The obtained cluster variables in NO ISR case are shown below:

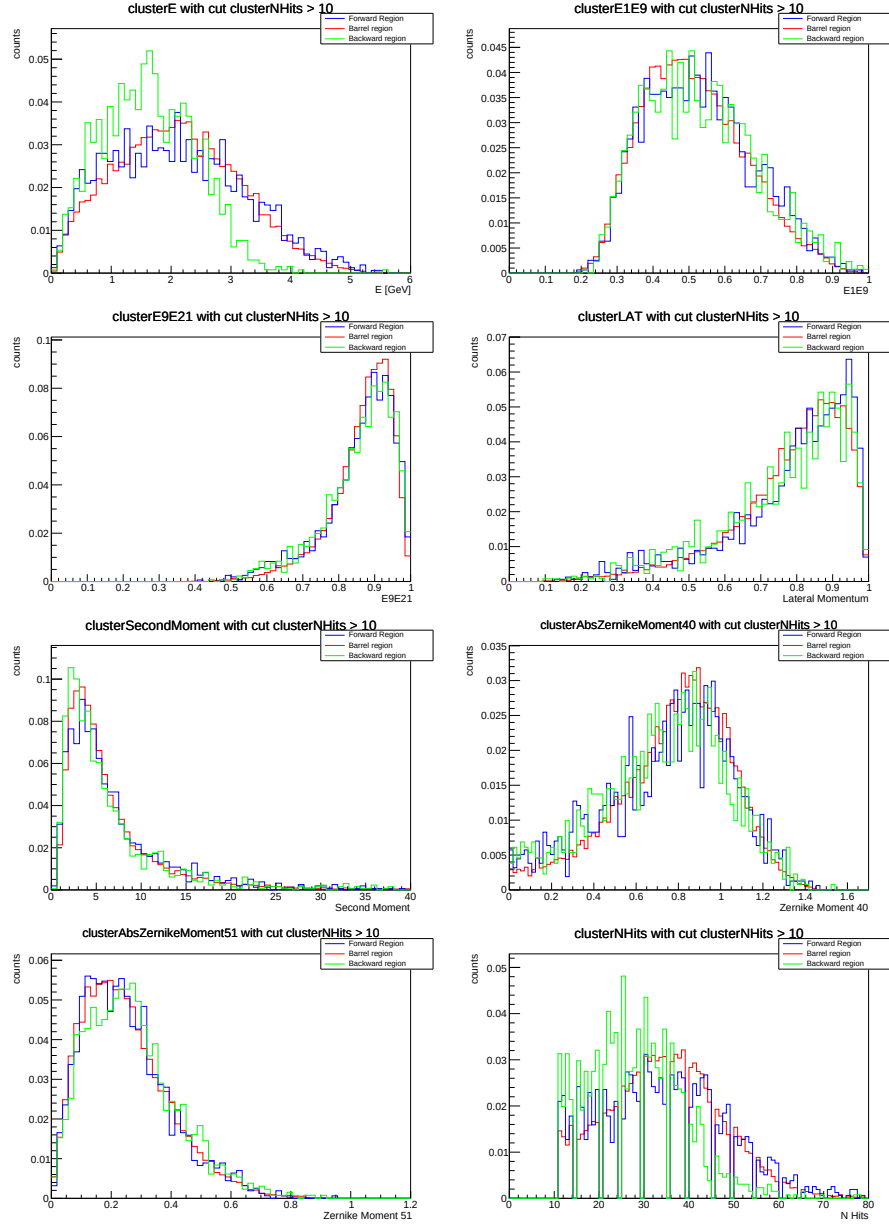


Figure 4.31: Comparison of reconstructed cluster variables for all $\bar{n}$ candidates in different ECL regions. The distributions are normalized to unit value

4.5 Summary of Signal Channel Analysis

The channel $e^+ + e^- \rightarrow p + \pi^- + \bar{n}$ has been studied, with both ISR and NO ISR emission.

The recoil system $p\pi^-(\gamma_{ISR})$ is correctly reconstructed by the analysis, as evidenced by the peak above the anti-neutron mass in the recoil mass distribution. Furthermore, the kinematic distributions (p and θ) of the recoil system are consistent with those of generator-level $\bar{n}$ candidates, in both ISR and NO ISR cases. At reconstructed level, good agreement is observed in the θ distribution, while the reconstructed $\bar{n}$ candidate momentum shows underestimation and poor resolution compared to the recoil system momentum, as observed in correlations and residuals distributions. Furthermore, a low-momentum value structure is observed in reconstructed and generated momentum of $\bar{n}$ candidates compared to the recoil system momentum.

Therefore, energy loss occurs within ECL clusters, suggesting leakage processes such as cluster split-off and/or pre-showering effects. Ancestor studies reveal that photons related to $\bar{n}$ particles are often misidentified as main $\bar{n}$ candidates, accounting for the low-momentum structure observed. In fact, the ancestors momentum can be partially recovered when good Monte Carlo associations occur, showing good agreement with the recoil system momentum as well. A study of $\bar{n}$ production reveals that candidates with $\bar{n}$ ancestors are generated far from the IP, before the ECL (pre-showering) and within the ECL (cluster split-off).

Furthermore, a one-constraint (1C) Kinematic Fit was tested, revealing no significant improvement in the reconstructed recoil momentum and confirming that the recoil system was already optimally reconstructed by the analysis.

Eventually, the reconstructed cluster variables associated with $\bar{n}$ candidates - such as cluster energy, lateral momentum, Zernike moments, etc.- were studied in the Monte Carlo signal sample. Similar structures to those observed in the momentum distributions were identified, particularly in the cluster energy and number of hits distributions. Therefore, ancestor studies confirm the momentum distribution observations: sub-clusters from secondary particles with $\bar{n}$ ancestry produce small-sized, low-energy structures in the cluster distributions.

In particular, a cut on the cluster number of hits yields reconstructed cluster variables closer to those expected from Monte Carlo $\bar{n}$ candidate selection, in both ISR and NO ISR cases.

5 Study of the Monte Carlo cocktail production

The second part of the analysis of this thesis is devoted to the study of **Belle II Monte Carlo cocktail simulations**. A Belle II "cocktail" MC production (typically generated with Pythia8) simulates all relevant physics channels simultaneously. Therefore, the analyzed channel $e^+e^- \rightarrow p\pi^-\bar{n}(\gamma_{\text{ISR}})$ is now embedded within all other background channels in the cocktail; the first task is to correctly reconstruct it amidst the surrounding backgrounds, studying the recoil mass distribution.

To achieve this, a first attempt uses the complete RUN1 $u\bar{u}$ production. Building on the signal sample discussed in the last chapter, further optimal cuts and selection criteria are developed to properly isolate the signal in the recoil mass distribution. Therefore the corresponding shower shape variables of neutral cluster associated to anti-neutron candidates are studied.

Eventually, the full Belle II $q\bar{q}$ ($u\bar{u}$, $d\bar{d}$, $c\bar{c}$, $s\bar{s}$) production is analyzed. The resulting cluster variables are compared with those from real data to assess Data/MC agreement.

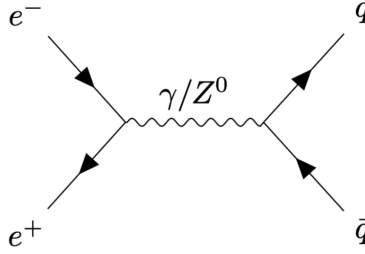


Figure 5.1: Tree level Feynman diagram of $e^+e^- \rightarrow q\bar{q}$

First of all, it could be useful start describing the Belle II Distributed Computing, namely the Belle II Grid.

5.1 Introduction to the Belle II Distributed Computing

Belle II datasets are categorized into two main types: **Data** and **Monte Carlo**. The Data are obtained from calibrated collision events processed by the experiment. Monte Carlo samples are generated using the Monte Carlo method according to various event generators, as discussed in Chapter 2. These are further classified into **Signal MC**- a specific process generated with a dedicated decay file upon analyst request (as in the previous chapter)- and **Generic MC**, or "cocktail", containing all processes expected in data with their correct relative fractions (as in the present chapter). These are usually in mdst (Mini Data Summary Table) format. There is cdst (Calibration Data Summary Table) as well, but it is used mainly to address calibration tasks.

Data and Monte Carlo samples can be "skimmed" by applying analysis-dependent

offline pre-selections called “skims”. Skims reduce data volume by selecting events of interest for specific analyses and retaining only the necessary information, producing smaller, more manageable datasets.

Using these Data and MC collections, analyses are performed via custom steering files with basf2, producing final ROOT ntuples. Subsequent offline analysis is then conducted outside the basf2 framework using ROOT, matplotlib, etc.

A scheme of the Belle II Data Model is shown below:

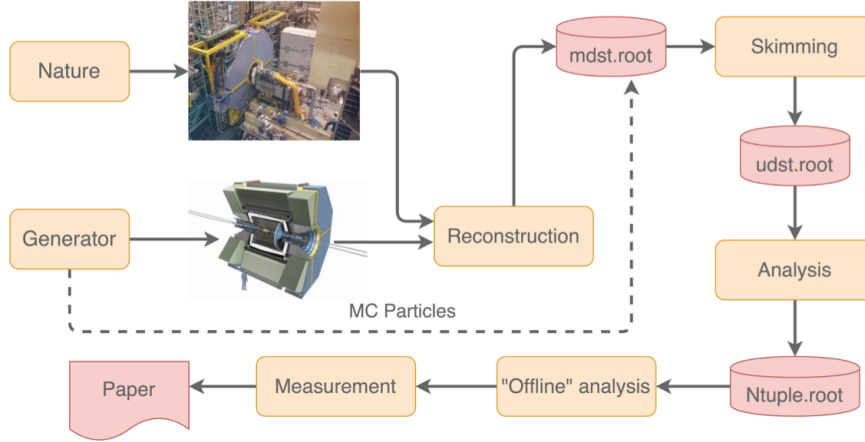


Figure 5.2: A scheme of the Belle II Data Model, starting from the nature processes information to the final results. As described, Monte Carlo truth information bypasses reconstruction, since the information is directly picked at generator level

Belle II Data and Monte Carlo samples are distributed across numerous storage sites worldwide. This Distributed Computing System, known as **the Grid**, consists of a network of loosely coupled computers with varying local conditions (operating systems, resources, etc.).

Two full copies of the raw data are maintained at dedicated data centers. Raw data is processed, skimmed, and distributed over other storage sites:

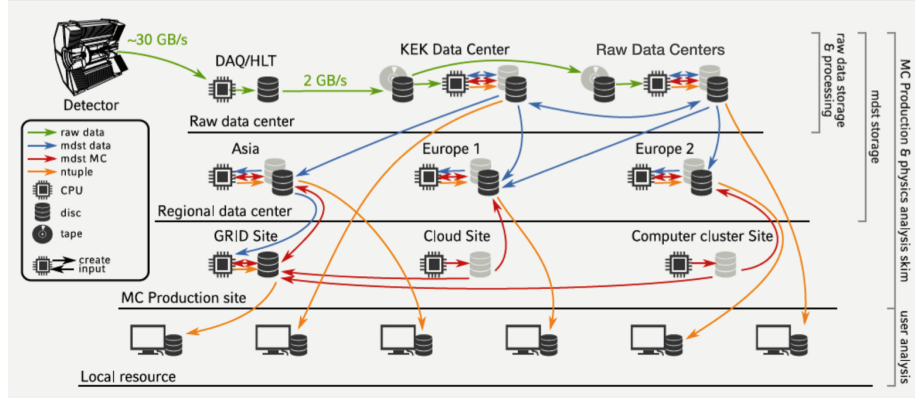


Figure 5.3: A scheme of the Belle II Distributed Computing (The Grid) sites

Each site interact with **DIRAC**, an interware that manages interactions between Belle II and its computing resources. **BelleDIRAC** is an extension that enables users to interact with DIRAC for job submission, data placement, monitoring, and other tasks.

5.1.1 Types of processes at the Belle experiment enviroment

Many processes can occur during an e^+e^- collision. These may mimic the selected signal and be accidentally reconstructed as signal events, though they represent background processes.

The types of processes present depend on if beams collide on-resonance or off-resonance in reference to the $\Upsilon(4S)$ resonance. $\Upsilon(4S)$ production occurs at $\sqrt{s} = 10.58$ GeV, yielding $\Upsilon(4S) \rightarrow BB$ processes. The B^+B^- and $B^0\bar{B}^0$ final states are referred to as charged and neutral $B\bar{B}$ pairs, respectively.

$e^+e^- \rightarrow q\bar{q}$ ($q = u, d, s, c$) processes, known as continuum production, also occur. The production cross section of the Belle II processes is shown below:

Process	Cross section (nb)
$e^+e^- \rightarrow \Upsilon(4S)$	1.10
$e^+e^- \rightarrow u\bar{u}(\gamma)$	1.61
$e^+e^- \rightarrow d\bar{d}(\gamma)$	0.40
$e^+e^- \rightarrow s\bar{s}(\gamma)$	0.38
$e^+e^- \rightarrow c\bar{c}(\gamma)$	1.30
$e^+e^- \rightarrow \tau^+\tau^-(\gamma)$	0.92
$e^+e^- \rightarrow e^+e^-(\gamma)$	300
$e^+e^- \rightarrow \mu^+\mu^-(\gamma)$	1.15
$e^+e^- \rightarrow \gamma\gamma(\gamma)$	4.99
$e^+e^- \rightarrow e^+e^-e^+e^-$	39.7
$e^+e^- \rightarrow e^+e^-\mu^+\mu^-$	18.9

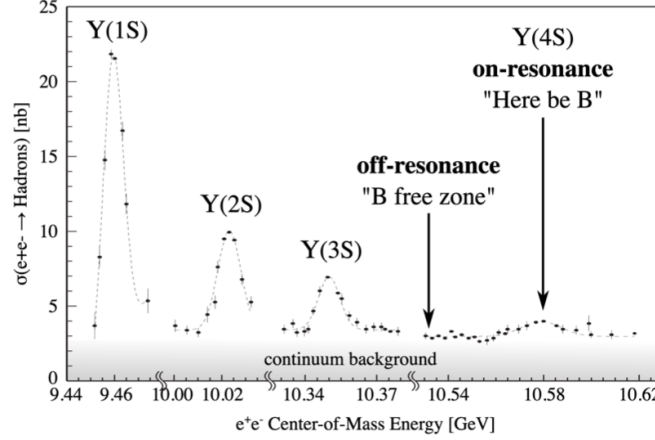


Figure 5.4: Cross section of $e^+e^- \rightarrow \text{Hadrons}$, in Υ resonances energy region [5]. $\Upsilon(1S)$ resonance is very narrow, while $\Upsilon(4S)$ one is broader and lower

5.2 Applied selections and cuts

To follow the same approach as used in the signal sample analysis, the recoil mass is first studied using the complete $u\bar{u}$ cocktail production, which corresponds to a $\sim 1429.23 \text{ fb}^{-1}$ integrated luminosity of data.

The following real cuts have been applied:

1. Recoil system: $0 \text{ GeV}/c^2 < \text{recoil mass} < 2 \text{ GeV}/c^2$
2. $\bar{n}$: neutral clusters from ECL and number of hits > 10 and energy $< 3 \text{ GeV}$
3. p : $\text{protonID} > 0.9$ and $dr < 1 \text{ cm}$ and $|dz| < 3 \text{ cm}$
4. π^- : $\text{pionID} > 0.1$ and $dr < 1 \text{ cm}$ and $|dz| < 3 \text{ cm}$
5. $\alpha < 0.20 \text{ rad} \sim 11.5^\circ$
6. No further charged particles in the Rest of Event (ROE)

Four additional cuts are applied beyond those used for the signal sample, such as cluster number of hits > 10 GeV, $\alpha < 0.20 \text{ rad}$, the request of no further charged particles in the Rest of Event (ROE) and the energy of the neutral cluster associated to $\bar{n}$ candidates must be less than 3 GeV.

The first cut arises from observations in the previous chapter; to achieve cleaner selection of neutral clusters associated with anti-neutron candidates -reducing cluster split-off and pre-showering contributes- a requirement on the number of lighted crystals is applied.

The second cut allows a stricter selection of anti-neutron candidates through a smaller α angle. α is defined as the 3D angle between the recoil vector and the

neutral cluster associated with the $\bar{n}$. As will be explained later, this threshold will be lowered further.

To explain the third cut, the concept of **Rest of Event** (ROE) must first be introduced. The ROE is built around the reconstructed target particles and consists of all other particles in the rest of event. In the present case, the reconstructed particles are $p, \pi^-, \bar{n}$ (and γ_{ISR}); therefore, the ROE includes all particles outside this reconstructed system. Consequently, the third selection criterion requires that no charged particles (i.e. tracks associated with charged particles) be present in the Rest Of Event, providing a cleaner event reconstruction.

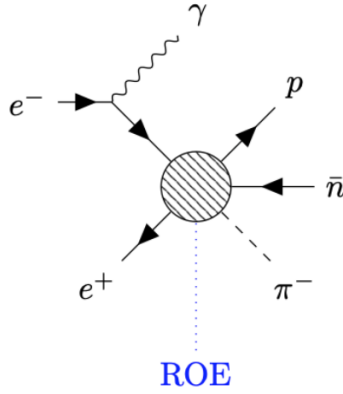


Figure 5.5: The selected channel with ROE made of a variable amount of particles.

The fourth cut arises from cluster energy studies performed on the first chunk of $u\bar{u}$ cocktail production, corresponding to $\sim 215 fb^{-1}$. The $\bar{n}$ list can be affected by several mis-identified γ_{ISR} . In order to study this phenomenon, the Monte Carlo ISR flag (mcISR) can be used to separate the two contributions in reconstructed $\bar{n}$ list. mcISR variable is a Monte Carlo flag; If mcISR = 0, the reconstructed particle is not an ISR photon. If mcISR = 1, the reconstructed particle is an ISR photon mis-identified as an anti-neutron. The obtained cluster energy distribution for $\bar{n}$ candidates is shown in figure 5.6.

It is easy to note that two structures at ~ 4 GeV and ~ 6 GeV, due to γ_{ISR} mis-identified as anti-neutrons, arise. In order to remove them, a cut on cluster energy such as those applied is mandatory.

However, γ radiation represents one of the most abundant produced particles in HEP experiments such as Belle II, making a major contribution to the background. In addition to the ISR ($E \sim 10$ MeV - 5 GeV) and FSR ($E < 10$ MeV) radiation described above, bremsstrahlung ($E < 50$ MeV) and secondary photons -such as those from $\pi^0 \rightarrow \gamma\gamma$ decays ($E < 400$ MeV)- are also present. Therefore, NO ISR channel where the recoil system is made of p, π^- should

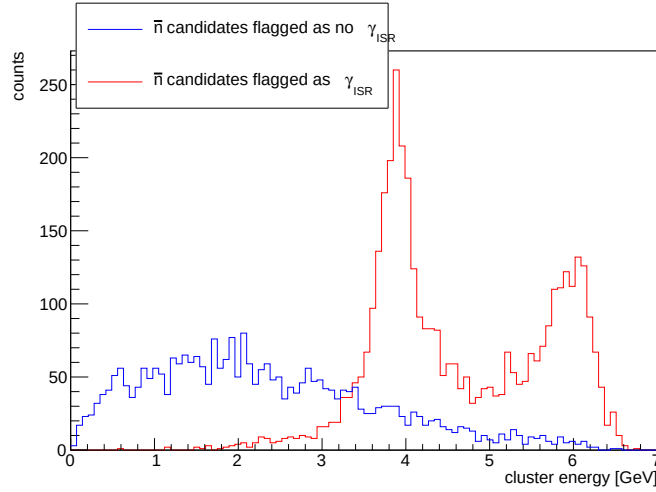


Figure 5.6: Cluster energy distribution of $\bar{n}$ candidates, divided by the Monte Carlo ISR flag (mcISR). If $\text{mcISR} = 0$, the reconstructed particle is not an ISR photon. If $\text{mcISR} = 1$, the reconstructed particle is an ISR photon. Of course this is a level-generator information.

represents a cleaner system to reconstruct than the ISR one where the recoil system is made of p, π^-, γ , since several background γ 's can be mis-identified as the ISR photon, which must be accounted for in the recoil system.

In order to compare the feasibility of selecting the channel in both ISR and NO ISR cases separately, the recoil mass distribution must be first studied using the same selection criteria. The better the selection of the signal from the background, the more consistent the shower shape variables will be for neutral clusters associated with $\bar{n}$ candidates, eventually allowing proper Data/MC agreement for these variables.

5.3 Recoil mass in $u\bar{u}$ cocktail production (NO ISR case)

The recoil mass distribution obtained from the $u\bar{u}$ cocktail is first studied in NO ISR case. If the recoil system p, π^- is properly reconstructed among all possible background channels, then a peak should be observed at the anti-neutron rest mass ($939.5 \text{ MeV}/c^2$). The obtained result is shown in 5.7.

A distinct structure is observed above the $\bar{n}$ mass value, suggesting that the selected channel is properly reconstructed. For a cleaner comparison between the signal channel and background components, the TopoAna output can be consulted 5.8. TopoAna does not account for particle-matter interactions; thus, all displayed particles originate directly from the e^+e^- collision at the IP.

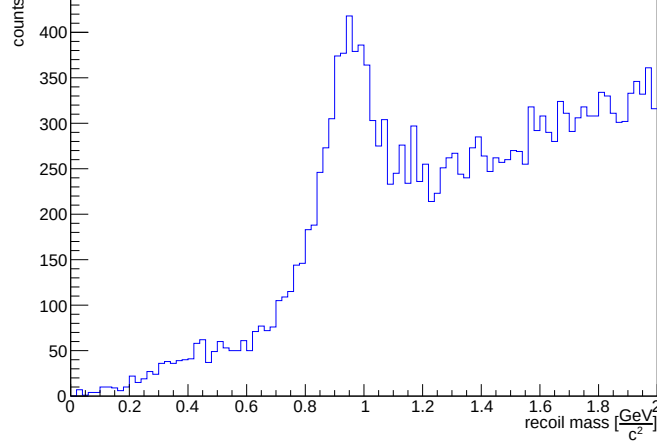


Figure 5.7: Reconstructed p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, NO ISR case.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow \pi^- \bar{n} p$	$\pi^- \bar{n} p$	24	2405	2405
2	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n} p$	$\pi^0 \pi^- \bar{n} p$	2	2317	4722
3	$e^+e^- \rightarrow \pi^- \bar{n} p \gamma^I$	$\pi^- \bar{n} p \gamma^I$	16	1980	6702
4	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n} p \gamma^I$	$\pi^0 \pi^- \bar{n} p \gamma^I$	6	1676	8378
5	$e^+e^- \rightarrow \pi^- \bar{n} p \gamma^I \gamma^I$	$\pi^- \bar{n} p \gamma^I \gamma^I$	12	751	9129
6	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n} p \gamma^I \gamma^I$	$\pi^0 \pi^- \bar{n} p \gamma^I \gamma^I$	3	612	9741
7	$e^+e^- \rightarrow \pi^+ \pi^- \gamma^I \gamma^I$	$\pi^+ \pi^- \gamma^I \gamma^I$	33	467	10208
8	$e^+e^- \rightarrow \pi^+ \pi^- \gamma^I$	$\pi^+ \pi^- \gamma^I$	0	381	10589
9	$e^+e^- \rightarrow \pi^- \bar{\Delta}^0 p, \bar{\Delta}^0 \rightarrow \pi^0 \bar{n}$	$\pi^0 \pi^- \bar{n} p$	28	313	10902
10	$e^+e^- \rightarrow \pi^0 p \bar{p}$	$\pi^0 p \bar{p}$	67	287	11189

Figure 5.8: First 10 channels of the TopoAna output in $u\bar{u}$ Monte Carlo cocktail, showing the different channel contributes, NO ISR case.

Accessing the Monte Carlo information from TopoAna, it is clear that the main contribution comes from the signal channel with $p\pi^-\bar{n}$ in the final state, immediately followed by the one with $p\pi^-\bar{n}\pi^0$ in the final state. Further contribution arise from $p\pi^-\bar{n}\gamma_{ISR}$, which could be interpreted as signal, even if, in NO ISR case, the γ_{ISR} is not explicitly reconstructed within the recoil system. The fourth contribute is due to the one with $p\pi^-\bar{n}\pi^0\gamma_{ISR}$ and so on. It is important to note that the contributions are sorted from most to least abundant.

5.3.1 Recoil mass components analysis

The recoil mass distribution components can be studied by correlating the ntuple information obtained from the steering file with that produced by the TopoAna program.

Each channel is identified by its own `iDcyTr` (index of the decay tree), which allows one to represent the contribution of a given channel (or a combination of channels) to a specific variable, such as the recoil mass of the system $p\pi^-$. The components are therefore denoted as follows:

1. $e^+e^- \rightarrow \pi^-\bar{n}p$ (`iDcyTr` == 24)
2. $e^+e^- \rightarrow \pi^0\pi^-\bar{n}p$ (`iDcyTr` == 2)
3. $e^+e^- \rightarrow \pi^-\bar{n}p\gamma_{ISR}$ (`iDcyTr` == 16)
4. $e^+e^- \rightarrow \pi^0\pi^-\bar{n}p\gamma_{ISR}$ (`iDcyTr` == 6)
5. $e^+e^- \rightarrow \text{other}$

The obtained stacked component to the recoil mass are shown in figure 5.9.

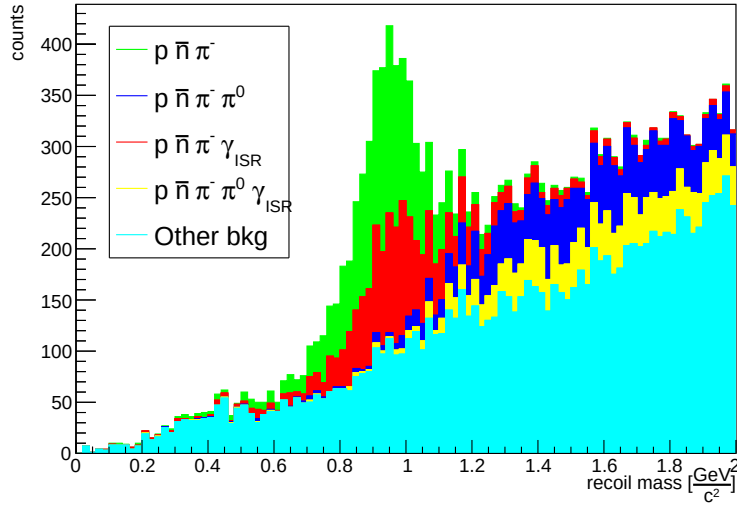


Figure 5.9: p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, displaying each stacked component contribution, NO ISR case.

Defining the *Signal Region* as the interval $[0.8, 1.2]$ GeV, it can be observed that contributions from the channel with `iDcyTr` = 2 predominantly lie to the right of the signal region. This is expected, as the recoil system computation excludes the π^0 . Same conclusion can be made for the with `iDcyTr` = 6. As expected, the signal channel (`iDcyTr` = 24) predominantly populates the signal region. The $\pi^-\bar{n}p\gamma_{ISR}$ channel (`iDcyTr` = 16) also mainly populates the signal region, indicating that the γ_{ISR} is negligible and has minimal impact on the recoil mass, as shown by the ISR energy distribution in Fig. 4.4. This can be directly observed from the single contribute from these two channels displayed in figure 5.10.

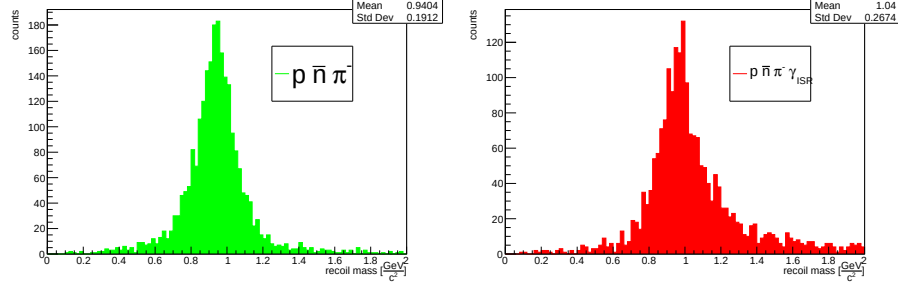


Figure 5.10: (a) p, π^- recoil mass and (b) p, π^-, γ_{ISR} recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail

Thus, despite the γ_{ISR} not being directly reconstructed, the $\pi^- \bar{n} p \gamma_{ISR}$ channel (iDcyTr = 16) can be considered a signal channel. This leads to the recoil mass composition shown in figure . 5.11.

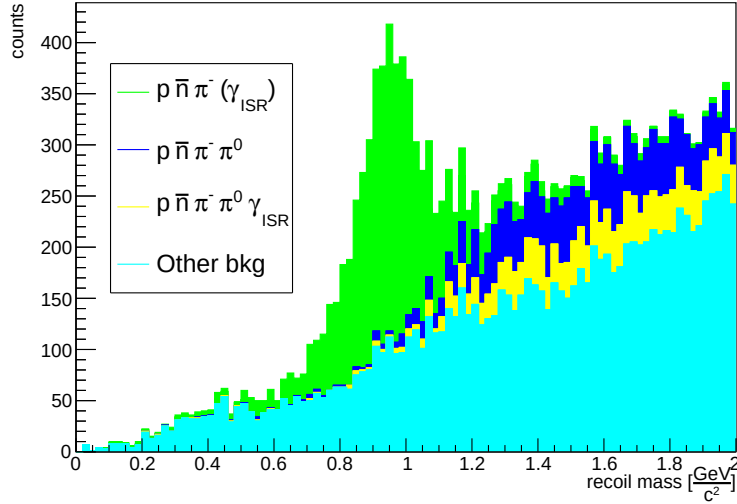


Figure 5.11: p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, displaying each stacked component contribution, NO ISR case.

The contributions from the remaining channels represent a flat background without discernible structures.

To further suppress the background, a tighter α cut can be studied. In the next subsection, the performance of the metrics and their usage are investigated in order to find the most appropriate cut in α .

5.3.2 Performance metrics study on α cut

In general, a **performance metric** quantifies how a classification process is optimized based on a specific parameter, which could, for example, be represented by a threshold.

In this specific HEP context, a performance metric quantifies how well the selection cuts distinguish the signal from the background. In this analysis, the signal can be assumed to be the channel with $\pi^- \bar{n} p(\gamma_{ISR})$ in the final state, while the background is represented by all the other channels.

It could be useful to investigate whether a stricter cut on the α angle -defined as the three-dimensional angle between the recoil system and the neutral cluster associated with the anti-neutron candidate- could improve the signal selection over the background.

To address this kind of analysis, performance metrics such as the **Figure of Merit** (FOM), the Signal Efficiency (ϵ_{sig}), the Background Efficiency (ϵ_{bkg}) and the **purity** can be employed. These are defined as follow:

- $$\text{FOM} = \frac{\text{SIGNAL}(\alpha)}{\sqrt{\text{SIGNAL}(\alpha) + \text{BACKGROUND}(\alpha)}},$$
- $$\epsilon_{sig} = \frac{\text{SIGNAL}(\alpha)}{\text{SIGNAL}(\alpha=0.20 \text{ rad})},$$
- $$\epsilon_{bkg} = \frac{\text{BACKGROUND}(\alpha)}{\text{BACKGROUND}(\alpha=0.20 \text{ rad})},$$
- $$\text{Purity} = \frac{\text{SIGNAL}(\alpha)}{\text{SIGNAL}(\alpha) + \text{BACKGROUND}(\alpha)},$$

where

- $\text{SIGNAL}(\alpha)$ is defined by the number of entries from the signal channel, depending on α .
- $\text{BACKGROUND}(\alpha)$ is defined by the number of entries from the other channels within the signal region, depending on α .

$\text{SIGNAL}(\alpha) + \text{BACKGROUND}(\alpha) = \text{TOTAL}(\alpha)$, the total amount of entries.

Assuming the signal region corresponds to the interval $[0.8, 1.2]$ GeV, the FOM characterizes the statistical significance of the applied threshold, namely the selected cut on α . The higher the FOM, the more signal is selected over the background (the figure is maximized). For each possible cut value of α , the Figure of Merit (FOM) is computed and the cut corresponding to the maximum FOM (or to a stable plateau) is selected. A higher FOM indicates a better trade-off between signal efficiency and background rejection, so the selection is optimized when the FOM is maximized. The obtained FOM is shown in Fig. 5.12.

One α value corresponding to the plateau can be chosen in order to maximize the FOM. The value $\alpha < 0.1$ rad ($\sim 5.73^\circ$) is therefore chosen, corresponding to $\text{FOM} = 43.4185$

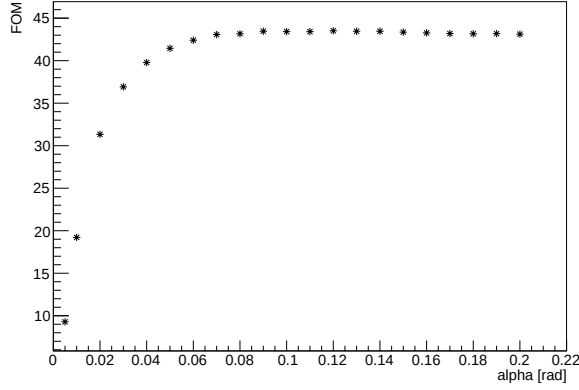


Figure 5.12: Figure of Merit (FOM) versus α angle for the $u\bar{u}$ cocktail.

Furthermore, the signal efficiency ϵ_{sig} and the background efficiency ϵ_{bkg} performance metrics can be adopted to study the Receiver Operating Characteristic (ROC), a curve that shows how well a binary classifier distinguishes signal from background as the decision threshold (α cut) is varied. Each point on the curve corresponds to a different threshold (a cut on the α angle). The Area Under the ROC Curve (AUC/ROC-AUC) is a single number summarizing the overall performance: 1.0 indicates perfect discrimination, while 0.5 corresponds to no better performance than random. The obtained ROC is shown in Fig. 5.13.

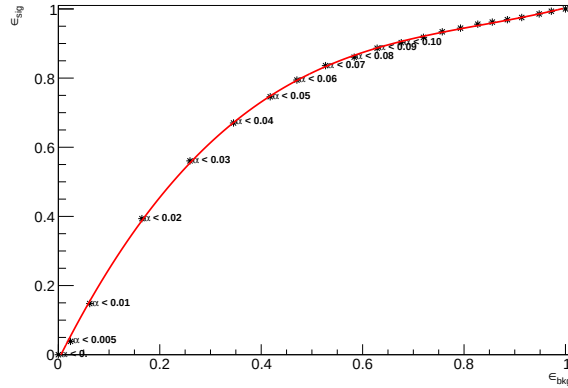


Figure 5.13: Receiver Operating Characteristic (ROC) curve constructed from the signal efficiency (ϵ_{sig}) and the Background Efficiency (ϵ_{bkg}) for the $u\bar{u}$ Monte Carlo cocktail. where each point corresponds to a different cut value on the α angle. For clarity, the labels are shown only for $\alpha < 0.10$ rad.

The Receiver Operating Characteristic (ROC) curve for the α angle cut yields an area under the curve (AUC) of 0.71, indicating good discrimination power between signal and background.

Therefore, the obtained performance metrics for $\alpha < 0.1$ rad are:

$$FOM = 43.4185, \epsilon_{sig} = 90\%, \epsilon_{bkg} = 68\%, Purity = 63\%$$

The obtained ϵ_{sig} vs $Purity$ is shown in Fig. 5.14.

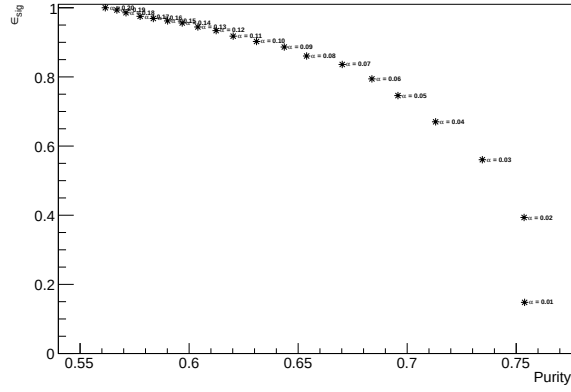


Figure 5.14: ϵ_{sig} vs $Purity$ curve for the $u\bar{u}$ Monte Carlo cocktail.

Following the optimization described, the recoil mass channel components distribution can now be analyzed using the cut $\alpha < 0.10$ rad.

The obtained stacked component to the recoil mass are shown in figure 5.15.

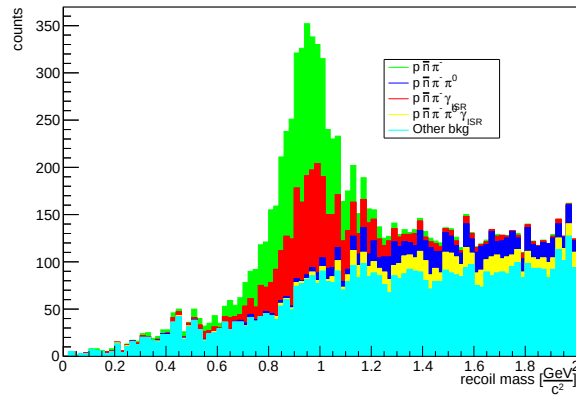


Figure 5.15: p, π^- recoil mass distribution in $u\bar{u}$ Monte Carlo cocktail, displaying each stacked component contribution, NO ISR case. $\alpha < 10$ rad cut is applied.

As expected, the signal is more prominent relative to the background than in the previous case. Background components containing π^0 peak outside the signal region $[0.8, 1.2]$ GeV on the right side, while other sources exhibit a lower, flat distribution across the recoil mass spectrum. The first 5 channels of $u\bar{u}$ TopoAna output with the $\alpha < 10$ rad cut is shown in 5.16.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow \pi^- \bar{n}p$	$\pi^- \bar{n}p$	14	2195	2195
2	$e^+e^- \rightarrow \pi^- \bar{n}p\gamma^I$	$\pi^- \bar{n}p\gamma^I$	10	1675	3870
3	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n}p$	$\pi^0 \pi^- \bar{n}p$	2	907	4777
4	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n}p\gamma^I$	$\pi^0 \pi^- \bar{n}p\gamma^I$	6	698	5475
5	$e^+e^- \rightarrow \pi^- \bar{n}p\gamma^I \gamma^I$	$\pi^- \bar{n}p\gamma^I \gamma^I$	9	598	6073

Figure 5.16: Top 5 channels from the TopoAna output for the $u\bar{u}$ Monte Carlo cocktails.

After applying the $\alpha < 0.10$ rad cut, the dominant contribution remains from the signal channel $p\pi^- \bar{n}$, now immediately followed by $p\pi^- \bar{n}\gamma_{ISR}$. As discussed, both contributions can be identified as signal since they fall within the signal region $[0.8, 1.2]$ GeV.

5.4 $q\bar{q}$ cocktail production (NO ISR case)

In order to fully characterize the Monte Carlo simulation, all $q\bar{q}$ continuum components must be studied. The Run 1 Monte Carlo productions for $u\bar{u}$, $d\bar{d}$, $s\bar{s}$, and $c\bar{c}$ are therefore analyzed. Each of them corresponds to an integrated luminosity of $\sim 1429.23 \text{ fb}^{-1}$ of data.

For a cleaner comparison between the signal channel and background components, the TopoAna output can be consulted in Fig. 5.17.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow \pi^- \bar{n}p$	$\pi^- \bar{n}p$	2	2794	2794
2	$e^+e^- \rightarrow \pi^- \bar{n}p\gamma^I$	$\pi^- \bar{n}p\gamma^I$	4	2163	4957
3	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n}p$	$\pi^0 \pi^- \bar{n}p$	1	1267	6224
4	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n}p\gamma^I$	$\pi^0 \pi^- \bar{n}p\gamma^I$	9	944	7168
5	$e^+e^- \rightarrow \pi^- \bar{n}p\gamma^I \gamma^I$	$\pi^- \bar{n}p\gamma^I \gamma^I$	5	767	7935
6	$e^+e^- \rightarrow \pi^+ \pi^- \gamma^I \gamma^I$	$\pi^+ \pi^- \gamma^I \gamma^I$	10	443	8378
7	$e^+e^- \rightarrow \pi^+ \pi^- \gamma^I$	$\pi^+ \pi^- \gamma^I$	7	424	8802
8	$e^+e^- \rightarrow \pi^0 \pi^- \bar{n}p\gamma^I \gamma^I$	$\pi^0 \pi^- \bar{n}p\gamma^I \gamma^I$	3	315	9117
9	$e^+e^- \rightarrow \pi^0 p\bar{p}$	$\pi^0 p\bar{p}$	98	244	9361
10	$e^+e^- \rightarrow K^- p\bar{\Lambda}, \bar{\Lambda} \rightarrow \pi^0 \bar{n}$	$\pi^0 K^- \bar{n}p$	226	216	9577

Figure 5.17: First 10 channels of the TopoAna output in $q\bar{q}$ Monte Carlo cocktail, showing the different channel contributes.

The TopoAna output reveals that the dominant contribution is from the signal channel $p\pi^- \bar{n}$ in the $q\bar{q}$ continuum. A secondary contribution comes from $p\pi^- \bar{n}\gamma_{ISR}$, which could be treated as signal despite the γ_{ISR} not being

explicitly reconstructed. The next contribution comes from $p\pi^-\bar{n}\pi^0$ in the final state. The fourth contribution is due to $p\pi^-\bar{n}\pi^0\gamma_{\text{ISR}}$, and so on. The situation is very similar to that observed in the $u\bar{u}$ case.

It would be interesting to discuss the individual TopoAna outputs separately for the $u\bar{u}$, $d\bar{d}$, $s\bar{s}$, and $c\bar{c}$ samples. The first 5 contributes for each sample are shown in 5.18.

rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow \pi^-\bar{n}p$	$\pi^-\bar{n}p$	14	2195	2195
2	$e^+e^- \rightarrow \pi^-\bar{n}p\gamma^I$	$\pi^-\bar{n}p\gamma^I$	10	1675	3870
3	$e^+e^- \rightarrow \pi^0\pi^-\bar{n}p$	$\pi^0\pi^-\bar{n}p$	2	907	4777
4	$e^+e^- \rightarrow \pi^0\pi^-\bar{n}p\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I$	6	698	5475
5	$e^+e^- \rightarrow \pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I$	9	598	6073
rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow \pi^-\bar{n}p$	$\pi^-\bar{n}p$	1	599	599
2	$e^+e^- \rightarrow \pi^-\bar{n}p\gamma^I$	$\pi^-\bar{n}p\gamma^I$	3	488	1087
3	$e^+e^- \rightarrow \pi^0\pi^-\bar{n}p$	$\pi^0\pi^-\bar{n}p$	0	360	1447
4	$e^+e^- \rightarrow \pi^0\pi^-\bar{n}p\gamma^I$	$\pi^0\pi^-\bar{n}p\gamma^I$	8	246	1693
5	$e^+e^- \rightarrow \pi^-\bar{n}p\gamma^I\gamma^I$	$\pi^-\bar{n}p\gamma^I\gamma^I$	4	169	1862
rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow K^-p\bar{\Lambda}, \bar{\Lambda} \rightarrow \pi^0\bar{n}$	$\pi^0K^-\bar{n}p$	3	120	120
2	$e^+e^- \rightarrow K^-p\bar{\Lambda}\gamma^I, \bar{\Lambda} \rightarrow \pi^0\bar{n}$	$\pi^0K^-\bar{n}p\gamma^I$	1	102	222
3	$e^+e^- \rightarrow K^0K^-\bar{n}p, K^0 \rightarrow K_L^0$	$K_L^0K^-\bar{n}p$	9	66	288
4	$e^+e^- \rightarrow \pi^0K^-p\bar{\Lambda}, \bar{\Lambda} \rightarrow \pi^0\bar{n}$	$\pi^0\pi^0K^-\bar{n}p$	15	52	340
5	$e^+e^- \rightarrow K^-p\bar{\Lambda}\gamma^I\gamma^I, \bar{\Lambda} \rightarrow \pi^0\bar{n}$	$\pi^0K^-\bar{n}p\gamma^I\gamma^I$	50	48	388
rowNo	decay tree	decay final state	iDcyTr	nEtr	nCEtr
1	$e^+e^- \rightarrow D^{*0}p\bar{\Lambda}_c^-\gamma^I, D^{*0} \rightarrow D^0\gamma, \bar{\Lambda}_c^- \rightarrow \pi^+\pi^-\bar{\Sigma}^-, D^0 \rightarrow \pi^0K_S^0, \bar{\Sigma}^- \rightarrow \pi^0\bar{p},$ $K_S^0 \rightarrow \pi^0\pi^0$	$\pi^0\pi^0\pi^0\pi^0\pi^+\pi^-\bar{p}\bar{p}\gamma^I\gamma$	0	1	1

Figure 5.18: Top 5 channels from the TopoAna output respectively for the $u\bar{u}$, $d\bar{d}$, $s\bar{s}$, and $c\bar{c}$ Monte Carlo cocktails (from top to bottom). The iDcyTr index values may differ across samples since each TopoAna output is produced independently from its corresponding ROOT ntuples.

A Signal sample analysis scripts

The scripts related to decay files, generation files and steering files are presented in the appendix A. Only the key highlights of the complete macros are reported, in order to ensure a smoother reading.

A.1 The decay file and the generation steering file

Monte Carlo (MC) truth is widely used in High Energy Physics (HEP) experiments. It allows the generation of datasets that follow a specific physical process, propagating particles through the detector and mimicking its behaviour layer by layer. These datasets are processed like real data during the reconstruction phase, with the advantage that the true information from the MC generator is known. The more realistic the MC simulation, the better the understanding of the real dataset.

The production of a MC sample is divided in three main steps:

- The generation of the MC particles with a generator
- The simulation of the detector structure and response to a particle (propagation)
- The reconstruction of tracks, cluster etc.

The **decay file** represents the first step to generate events specifying the decay channels of interest. It is, as the name suggests, the script in which the decay channel of a specific particle is explicitly set to a desired mode. It allows to impose several constraints on particle decay modes, in order to study a specific detector response or the kinematics of a given particle. This is called in jargon "generating signal MC events". The related file has *.dec* extension. An example from section 3 is explained below:

```
1 Decay vpho
2 1.0 p+ anti-n0 pi- PHSP;
3 Enddecay
4
5 End
```

Figure A.1: Decay file example powered by Evt_Gen

The declaration of a decay begins with the keyword *Decay* and ends with *Enddecay*. In this case, the dummy particle *vpho* is constrained to decay into a proton (p+), an antineutron (anti-n0) and a negative pion (pi-) according to the Phase Space Model (PHSP), with branching ratio set to 1.0. Since the channel is the following one

$$e^+ + e^- \rightarrow p + \pi^- + \bar{n} (+ \gamma_{ISR})$$

in order to produce this specific channel with ISR emission, both *PHOKHARA* and *Evt_Gen* generators are employed. The first one is acting as ISR generator by generating:

$$e^+ + e^- \rightarrow vpho (+ \gamma_{ISR})$$

Finally, the virtual photon is decayed by *Evt_Gen*, according to the shown above decay file:

$$vpho \rightarrow p + \pi^- + \bar{n}$$

To assign a particle the same final states as its charge-conjugated partner (assuming charge conjugation), the *CDecay* identifier can be used, followed by the charge-conjugated state. However it is not employed in the present case. The **generation steering file** is the operative script that generates a MC sample. An example is given below:

```

9  main = b2.Path()
10
11 # Define number of events (-n number -o output in terminal) and experiment number
12 main.add_module('EventInfoSetter', explist=[0])
13
14 print("***** CHANNEL e+ e- (isr) --> vpho --> p+ pi- nbar ***** ")
15 final_state = ['p+', 'pi-', 'anti-n0']
16 decfile = "../dec_file/dec_isr.dec"
17 #Generate isr events with phokhara_evt_gen
18 ge.add_phokhara_evtgen_combination(path=main, final_state_particles = final_state, user_decay_file = decfile,)
19
20
21 main.add_module('PrintMCParticles', onlyPrimaries = False)
22
23 # Simulate the detector response
24 si.add_simulation(path=main)
25
26 # Reconstruct the objects
27 re.add_reconstruction(path=main)
28
29 mdst.add_mdst_output(path=main)
30 # Process the steering path
31 b2.process(path=main)
32
33 # Finally, print out some statistics about the modules execution
34 print(b2.statistics)

```

Figure A.2: Usage of PHOKHARA&Evt_Gen, where simulation and reconstruction are performed

As previously discussed, basf2 scripts are structured as modules, which are sequentially arranged within a path, which is called *main* in this example (row 9). The first module (row 12) sets the desired amount of events and the specific experiment number, which is provided via keyboard input prior to script interpretation. Then, the signal from the decay file is added (row 18), with **add_phokhara_evtgen_combination** from generators basf2 modules ¹. From row 24 to 27, the simulation and the reconstruction, as described in section 3.3.1, are added to the path. Then the output file is created to store the MC truth, namely the generated event with a specific signal sample. Finally the steering path is processed in row 31.

¹<https://software.belle2.org/development/sphinx/generators/doc/index.html>

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